

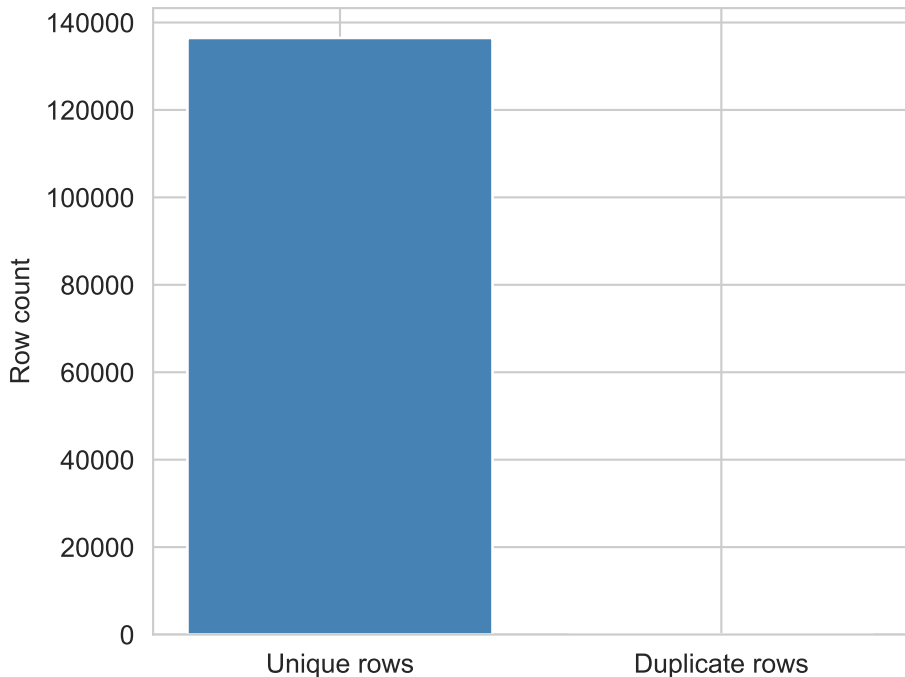
Solar AC Power Forecasting

48h -> 15min Sequence Forecasting: EDA, Modeling & Comparison

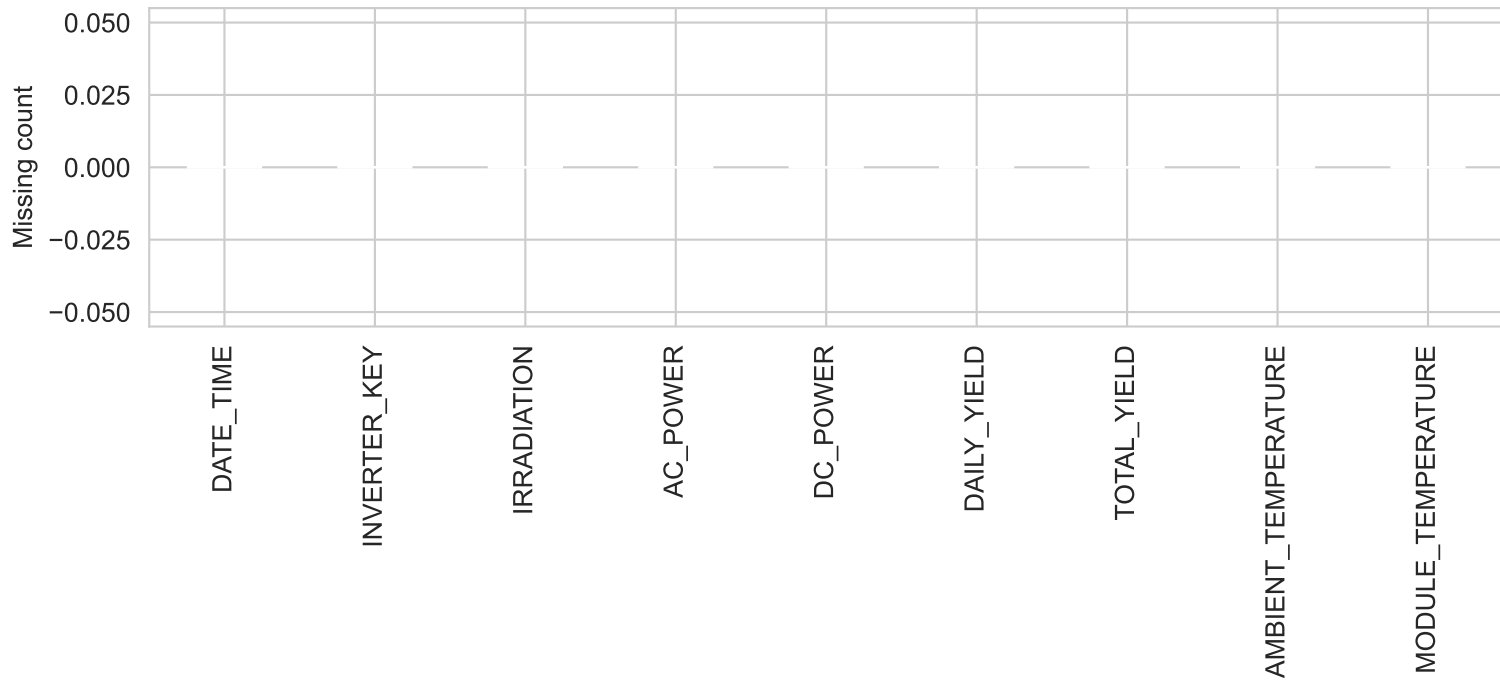
Contents:

1. Data Quality Checks
2. Exploratory Data Analysis (EDA)
3. Cleaning: Night Adjustment, Fault Isolation, Invalid Values
4. Feature Engineering: MODULE_DERATING_FACTOR
5. Time-Series Segmentation & Sequence Generation
6. Linear Regression / Random Forest / XGBoost / LightGBM / LSTM
7. Model Comparison & Diagnostics
8. Best Model Conclusion

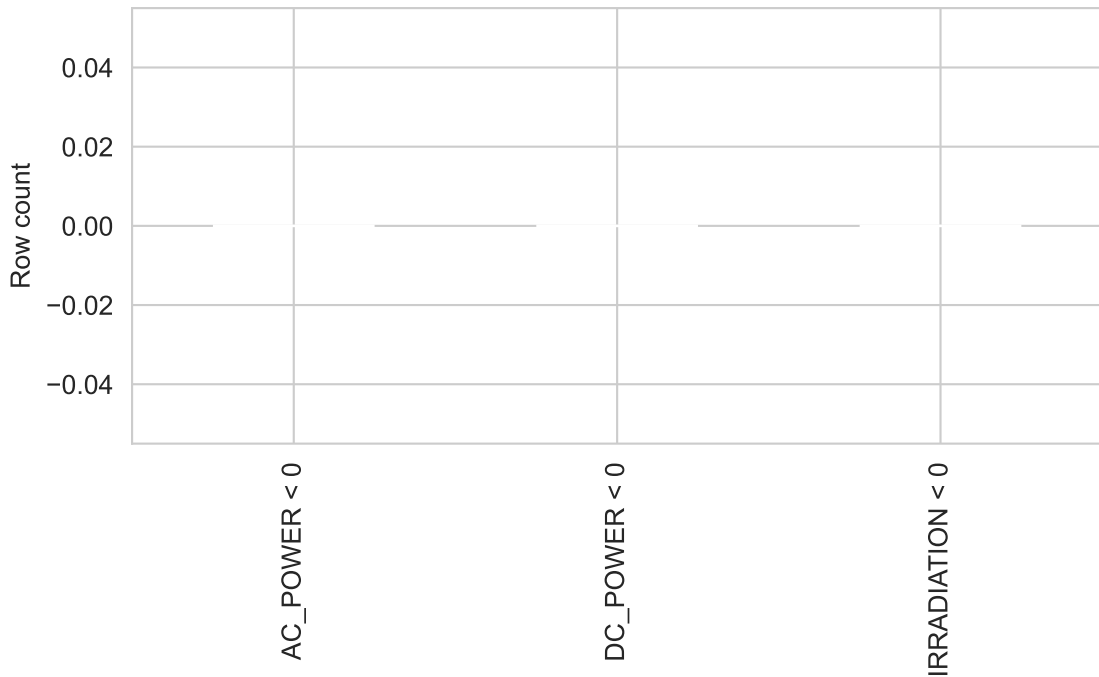
Duplicate Rows



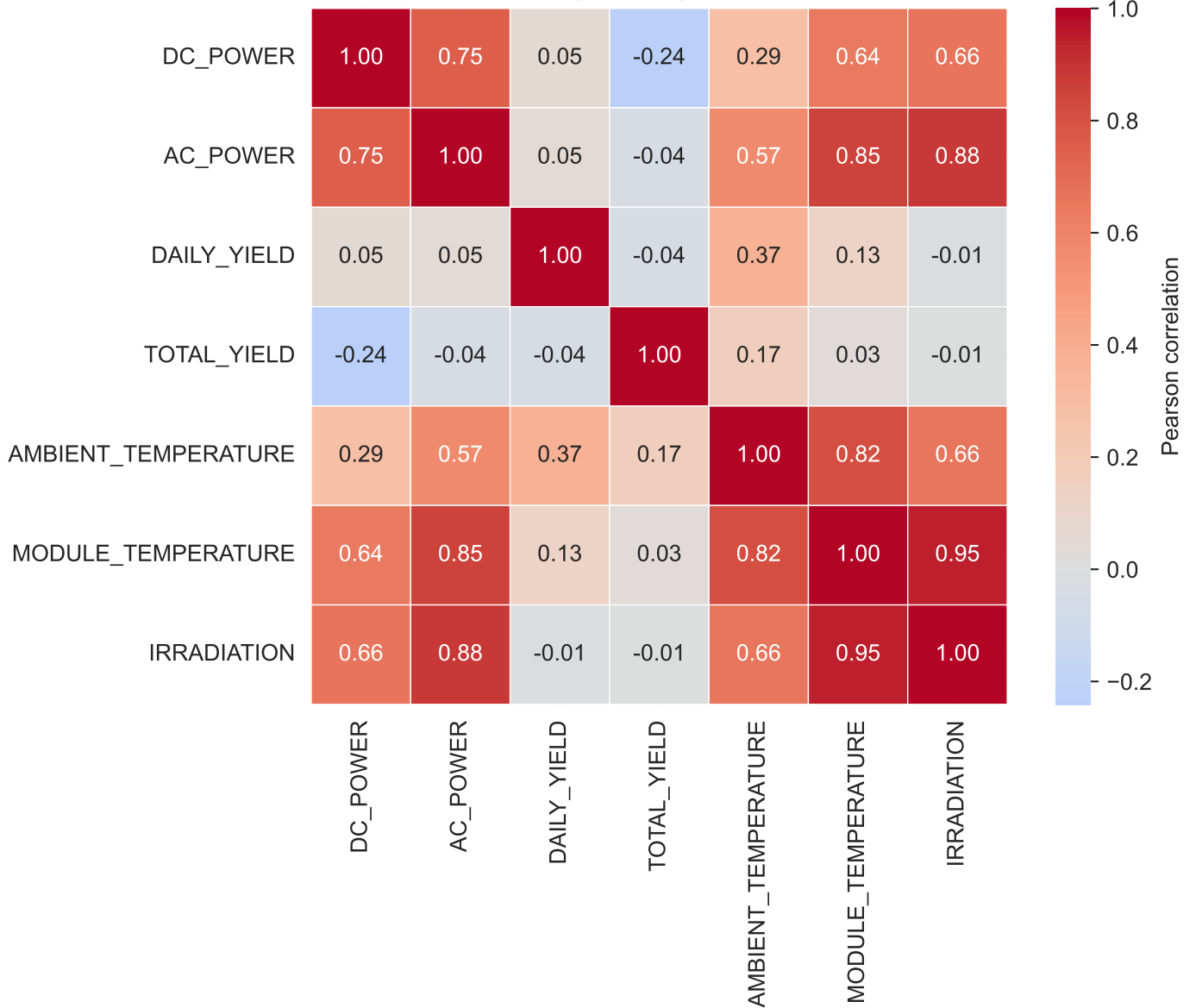
Missing Values per Required Column



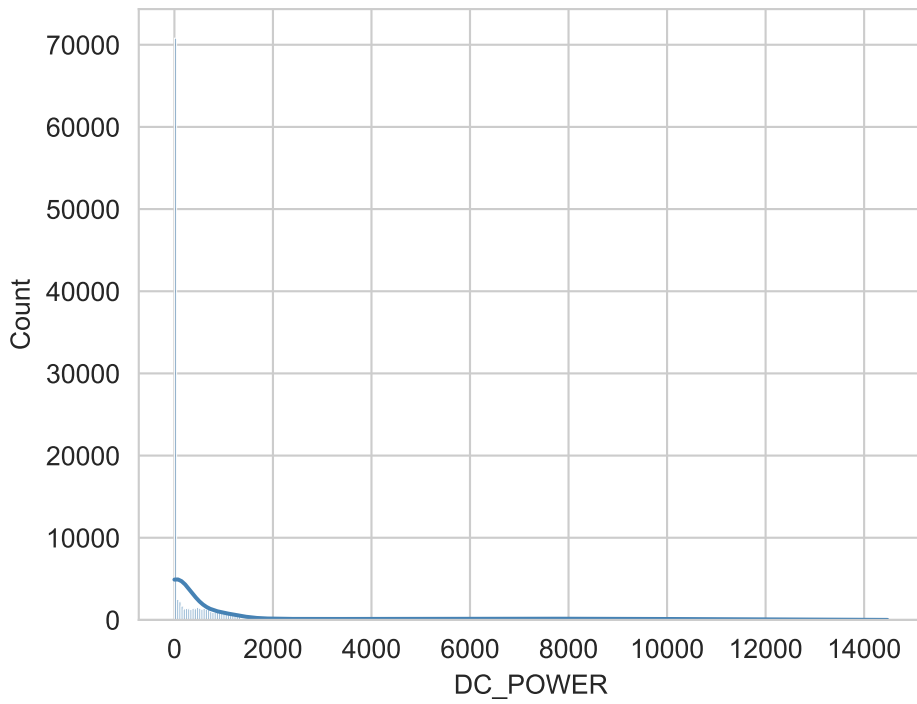
Negative Values by Column



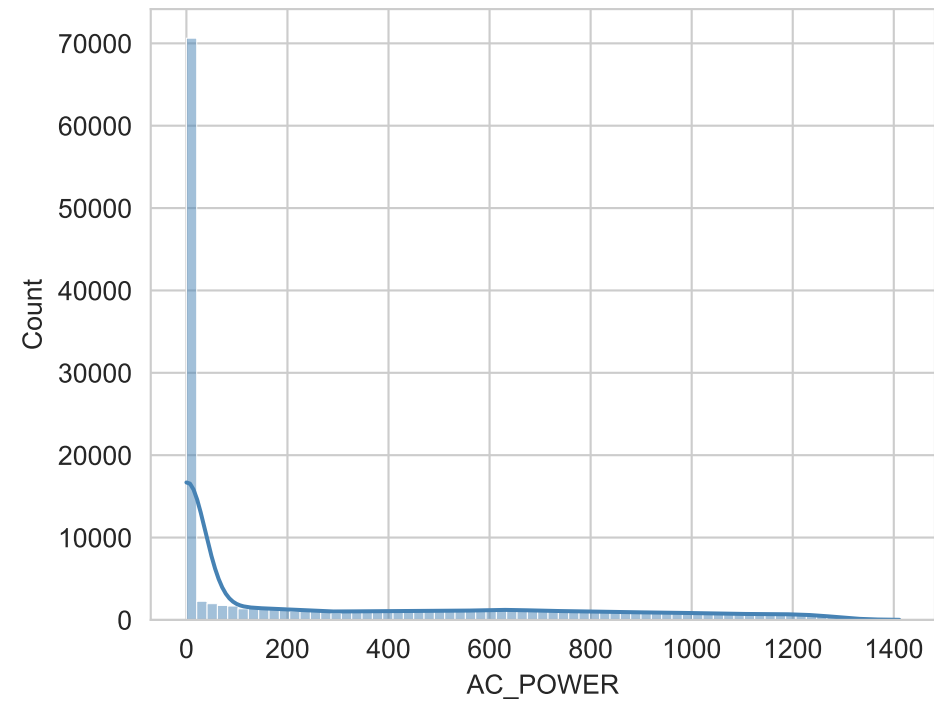
Correlation Heatmap — Sequence Feature Columns



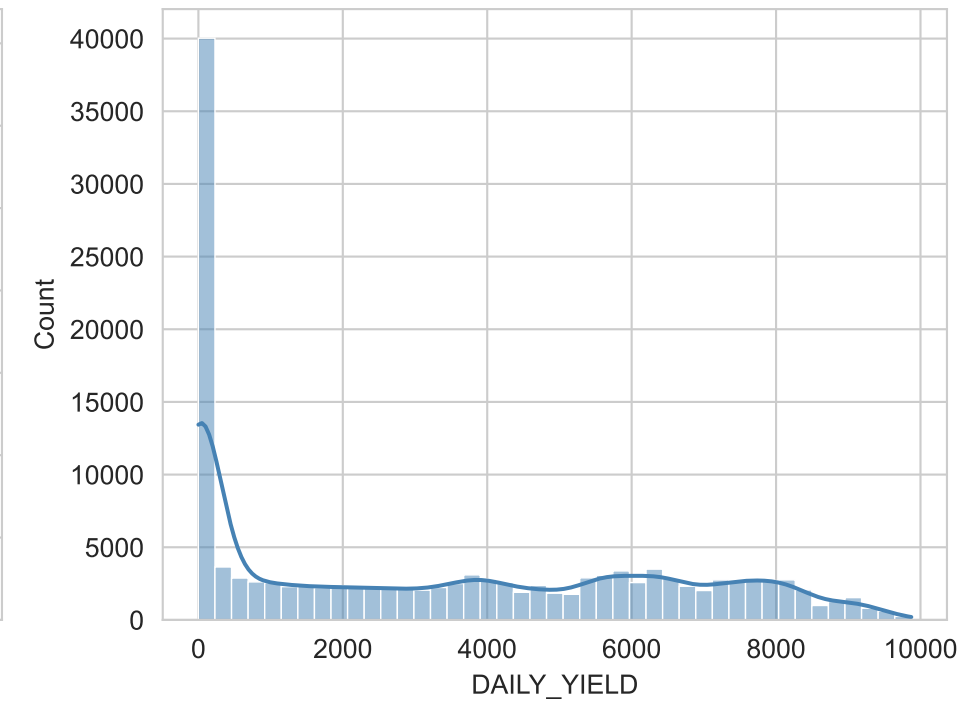
Distribution of DC_POWER



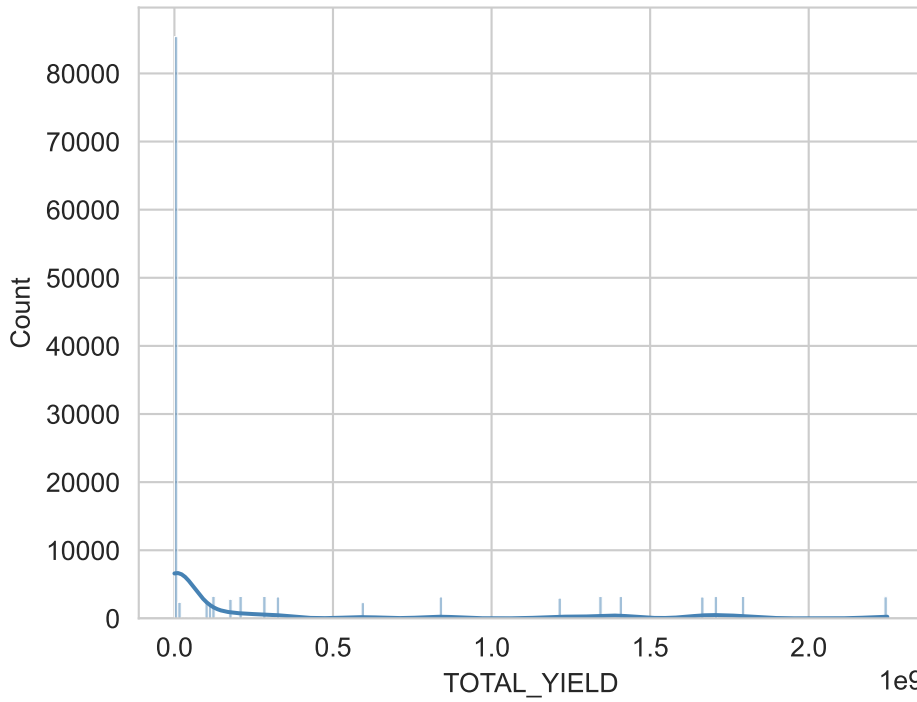
Distribution of AC_POWER



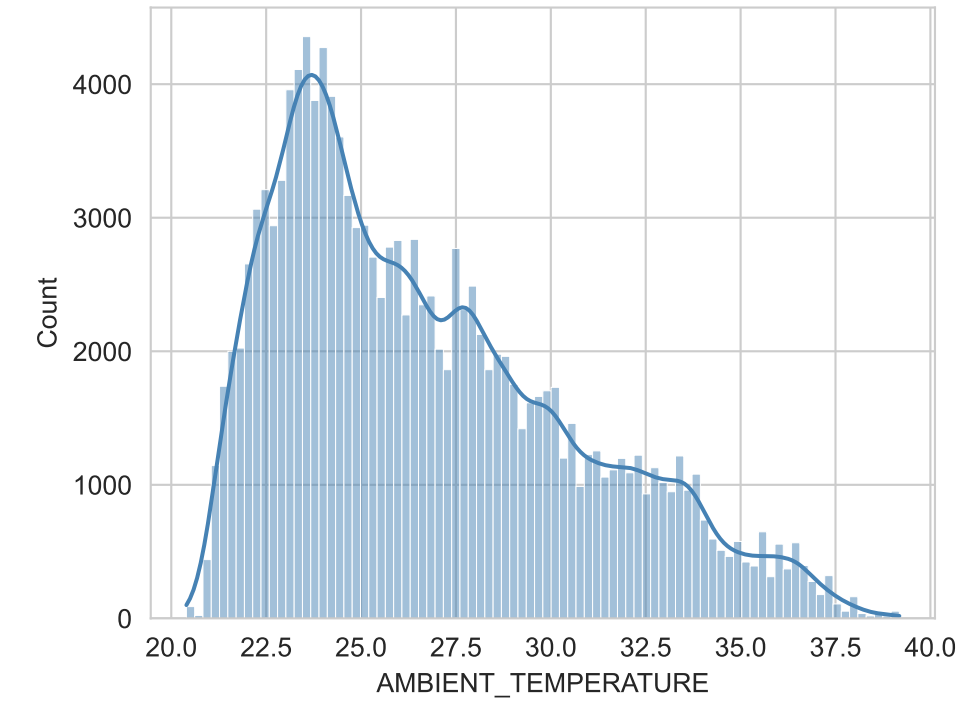
Distribution of DAILY_YIELD



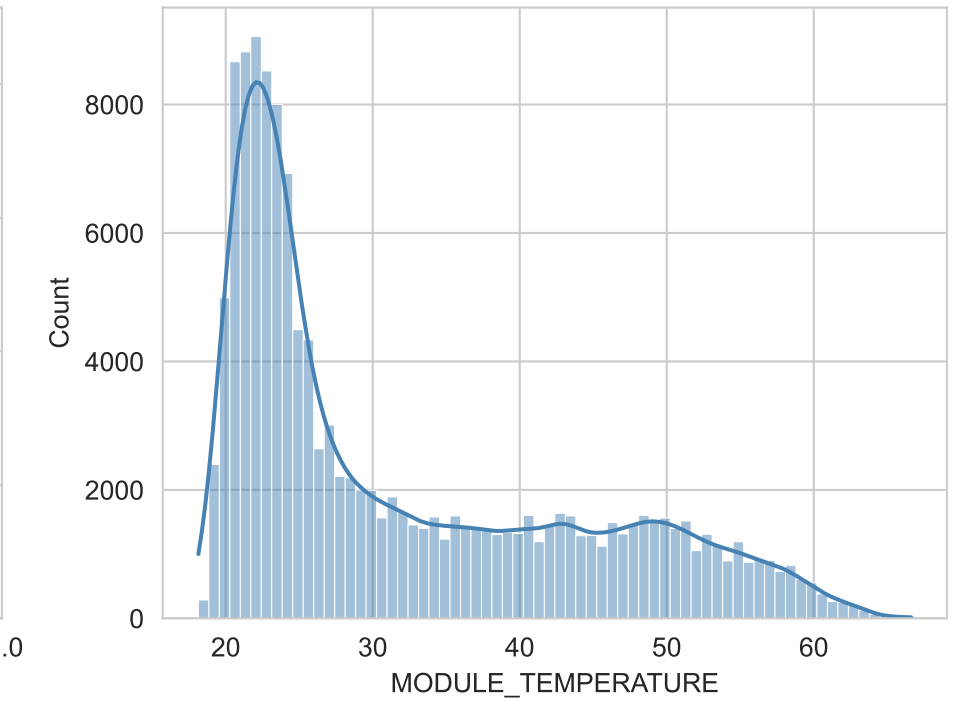
Distribution of TOTAL_YIELD



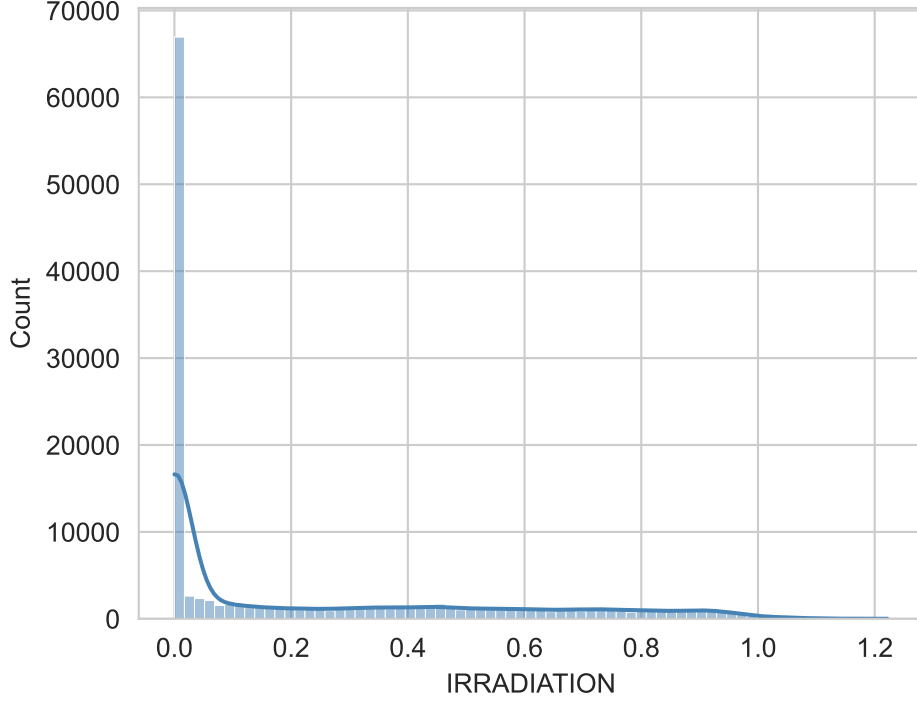
Distribution of AMBIENT_TEMPERATURE



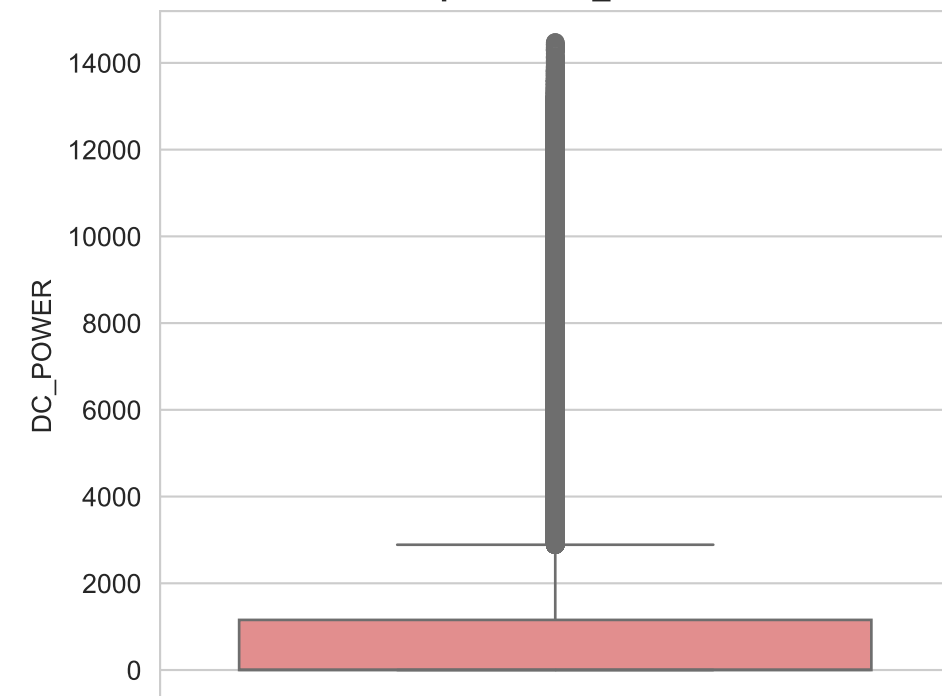
Distribution of MODULE_TEMPERATURE



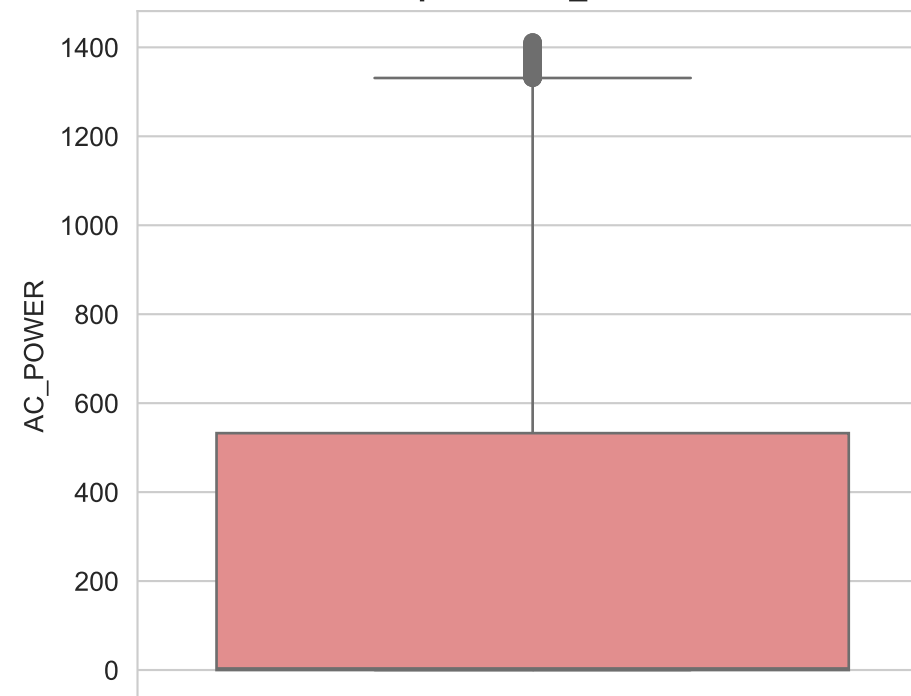
Distribution of IRRADIATION



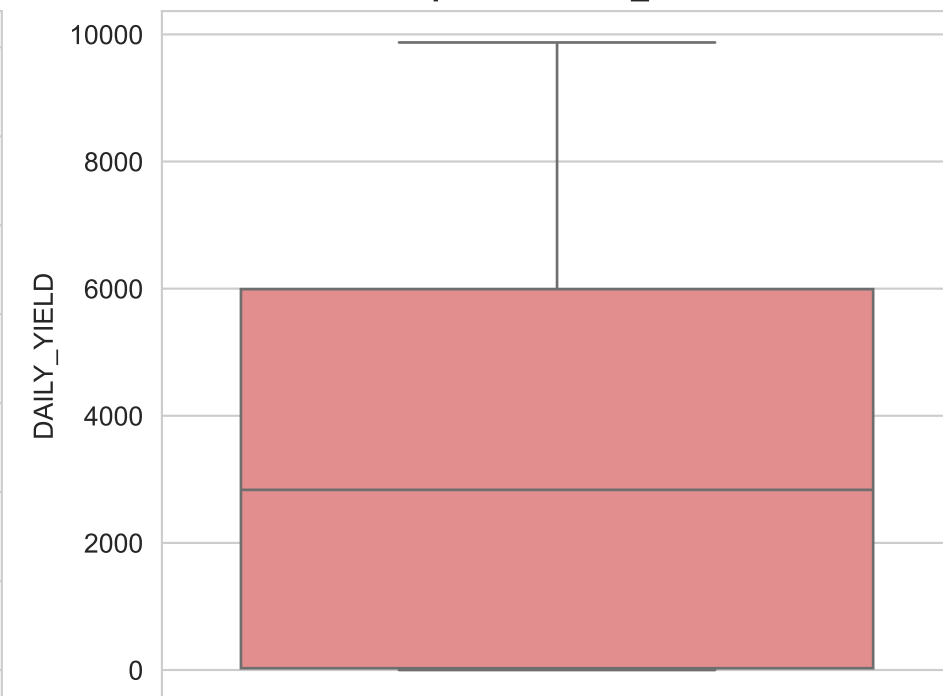
Boxplot of DC_POWER



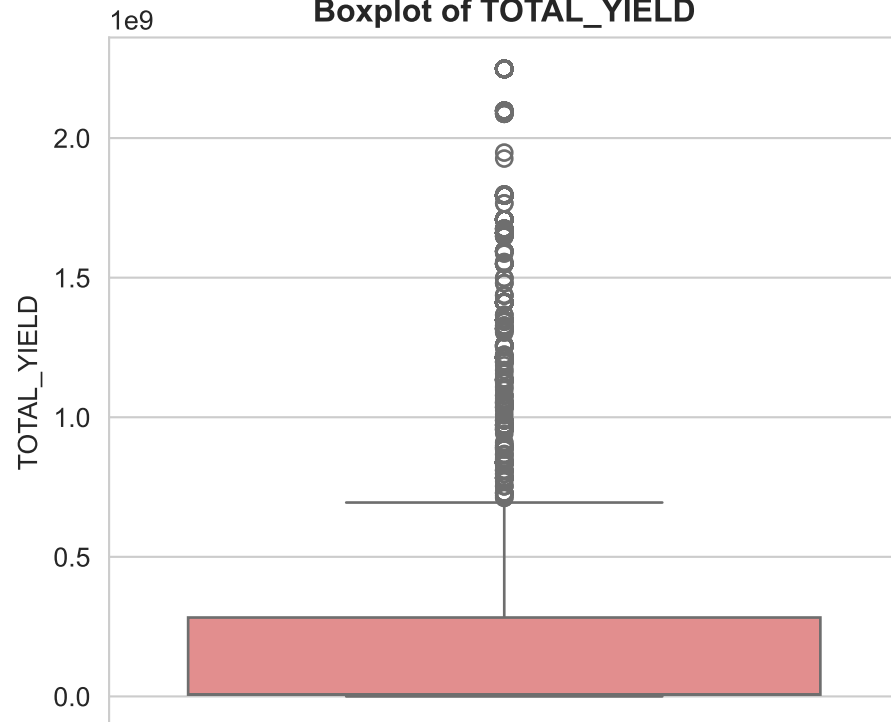
Boxplot of AC_POWER



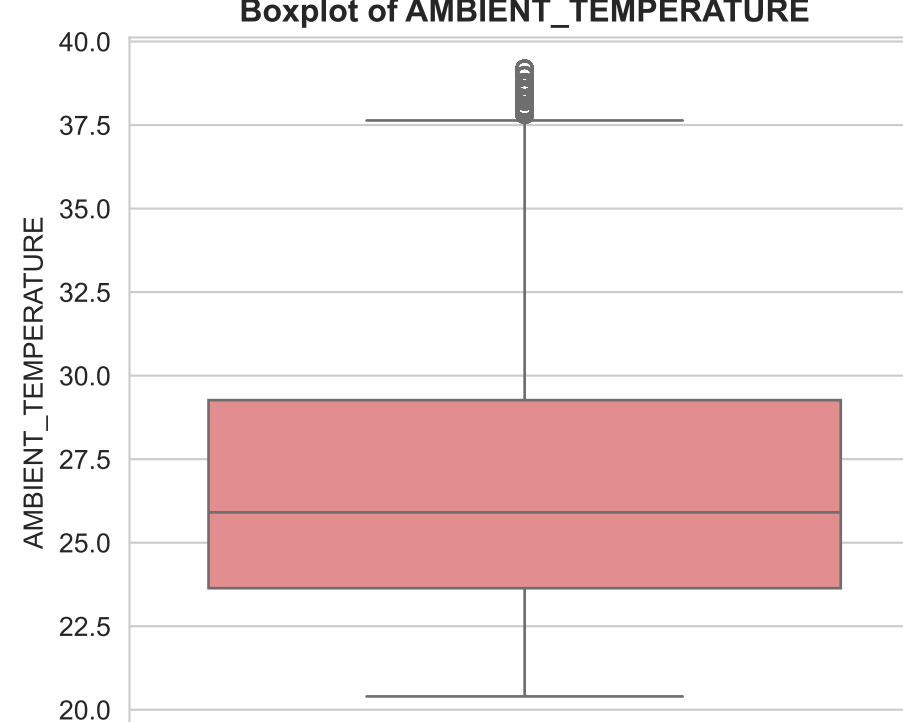
Boxplot of DAILY_YIELD



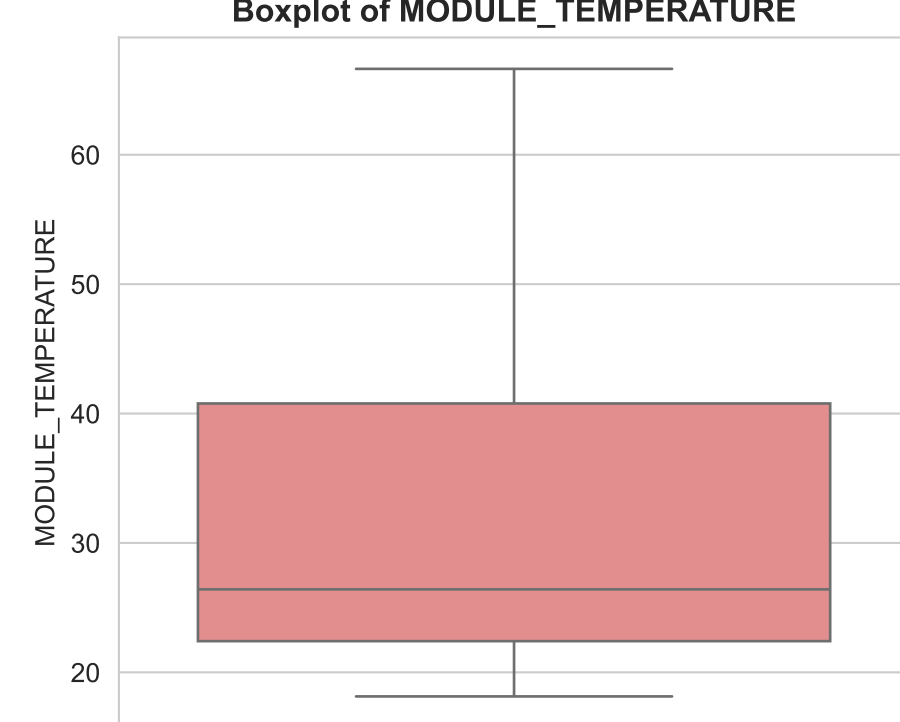
Boxplot of TOTAL_YIELD



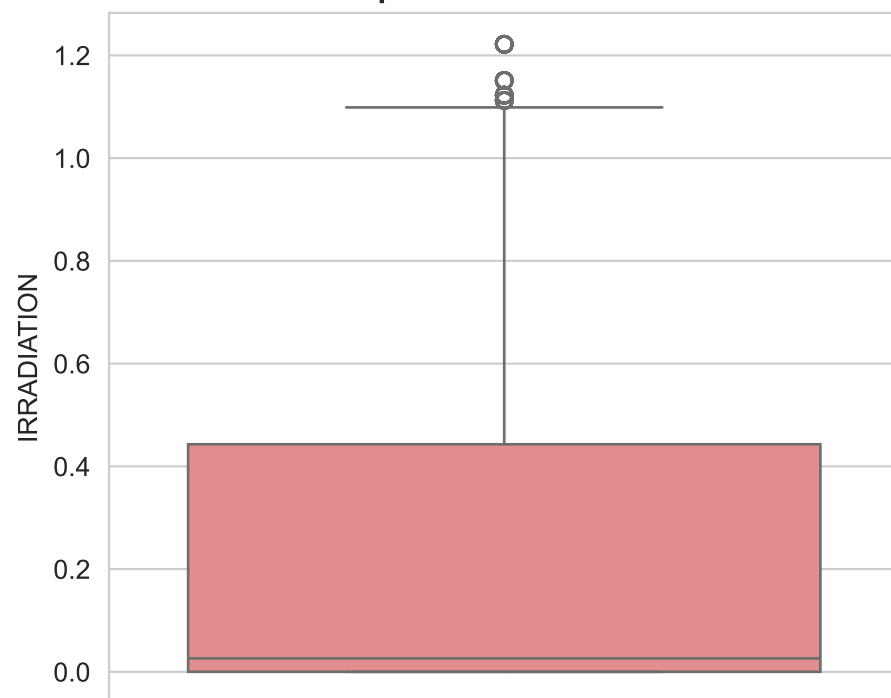
Boxplot of AMBIENT_TEMPERATURE

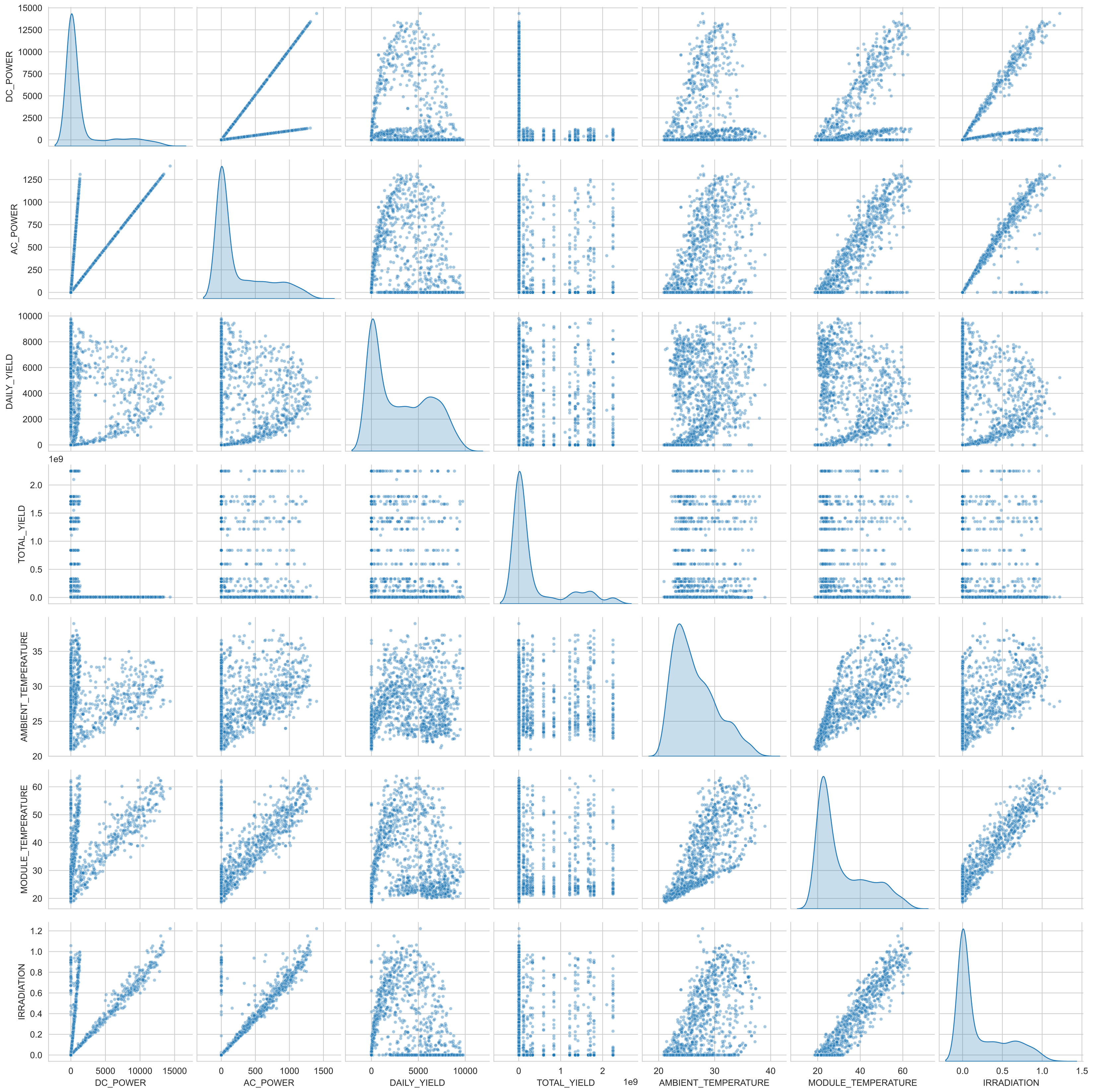


Boxplot of MODULE_TEMPERATURE

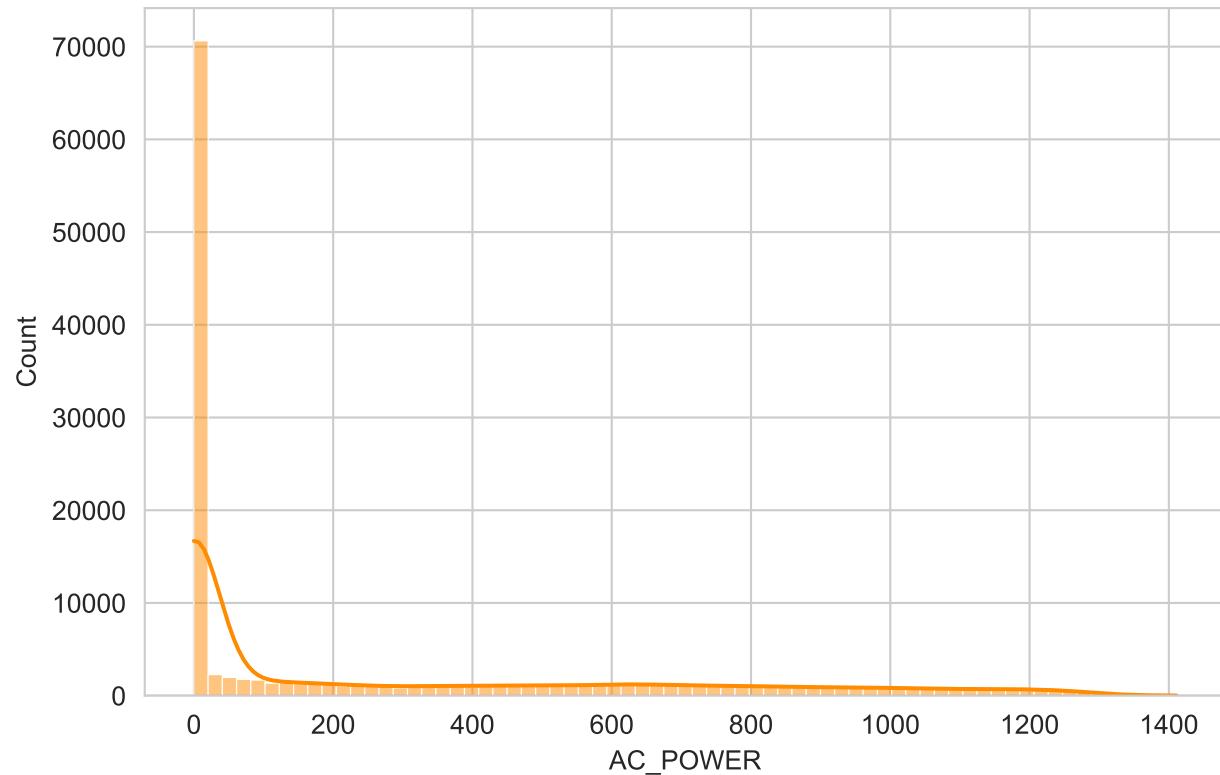


Boxplot of IRRADIATION

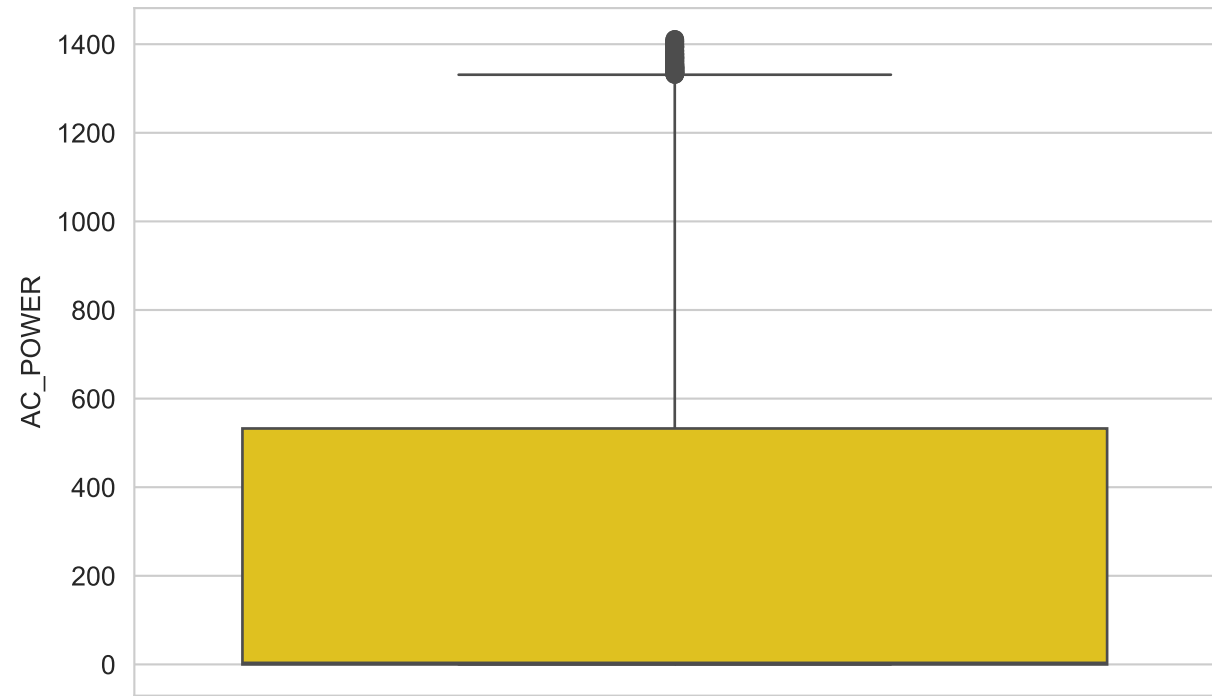




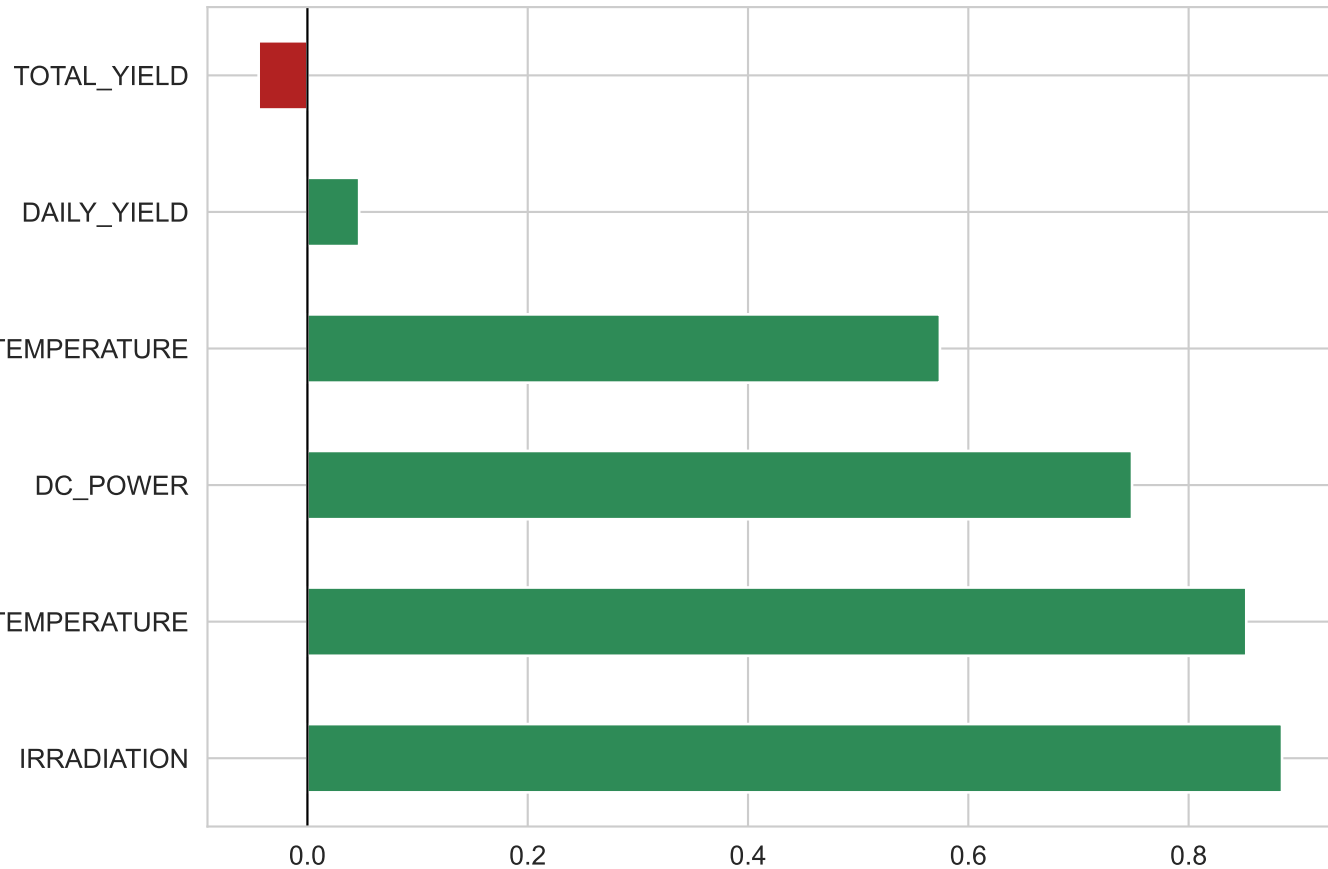
Distribution of AC_POWER



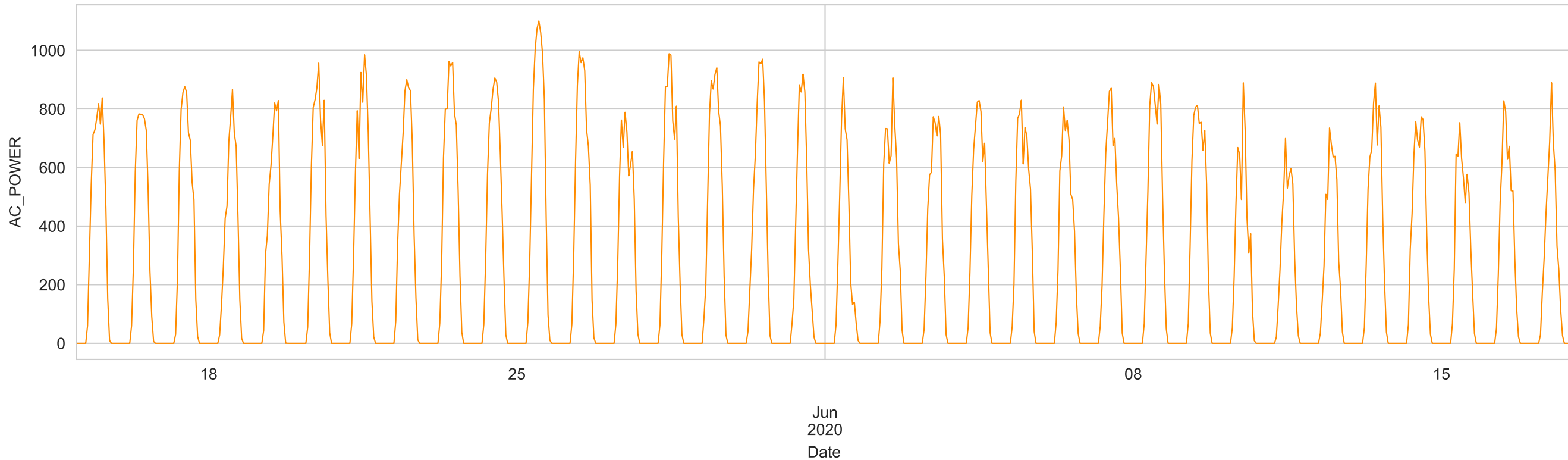
Boxplot of AC_POWER



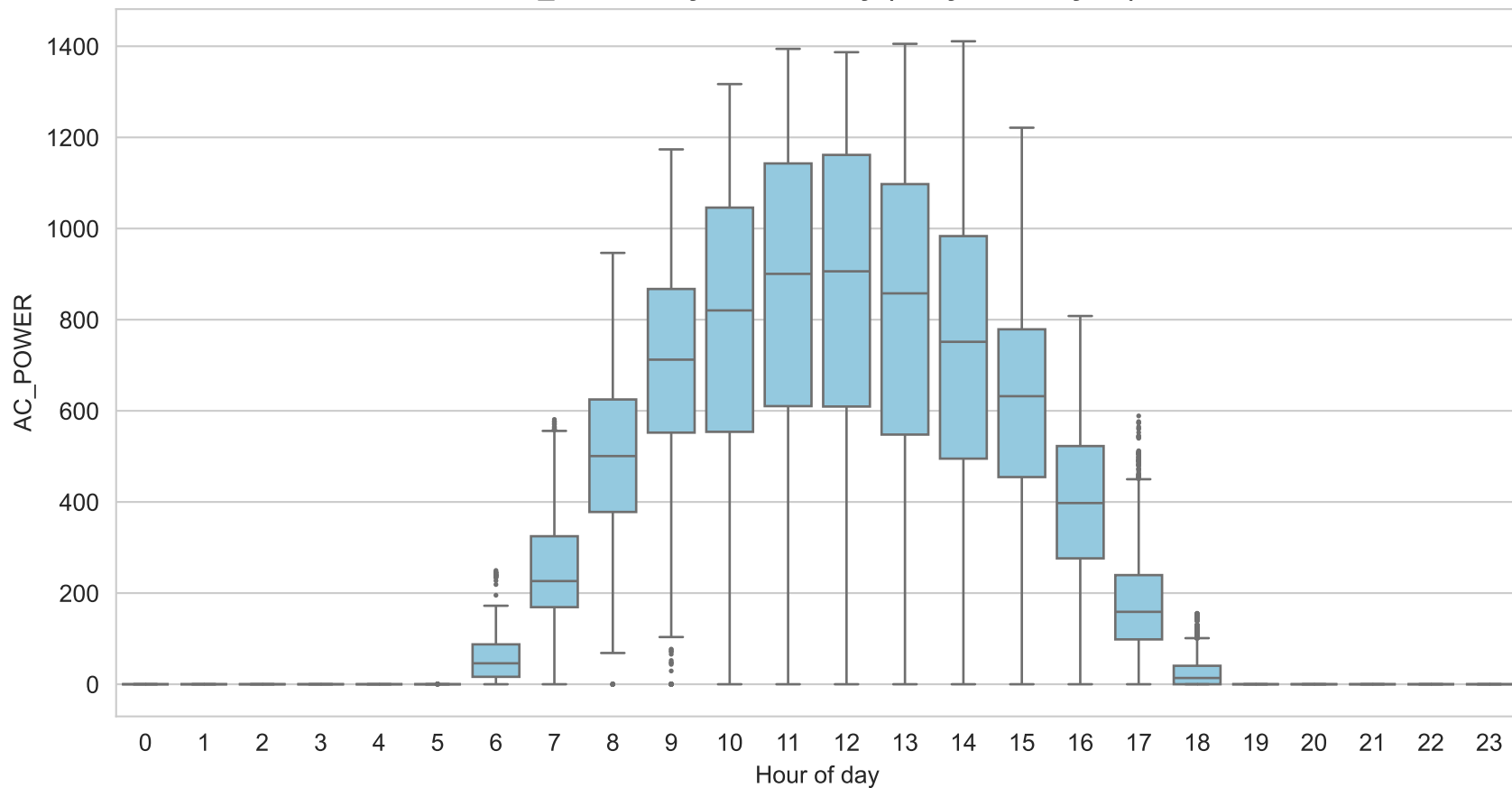
Feature Correlation with AC_POWER



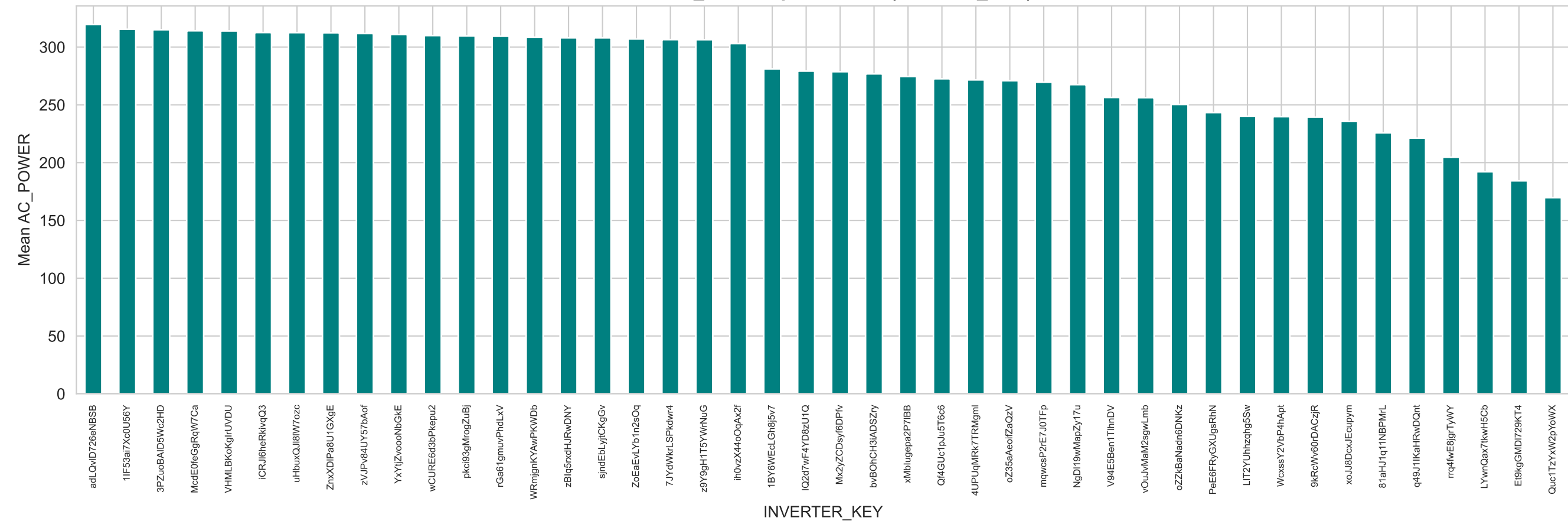
Fleet-Average AC_POWER Over Time (hourly mean)



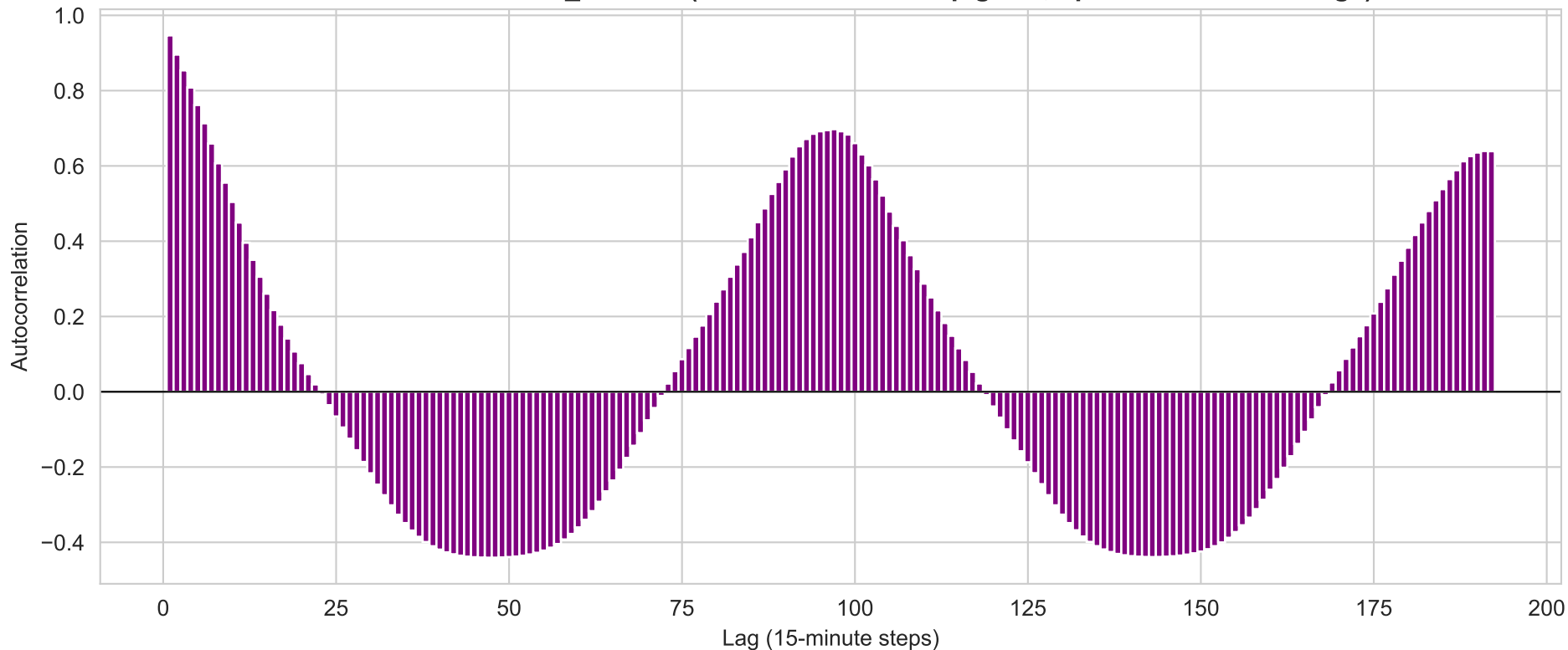
AC_POWER by Hour of Day (Daily Solar Cycle)



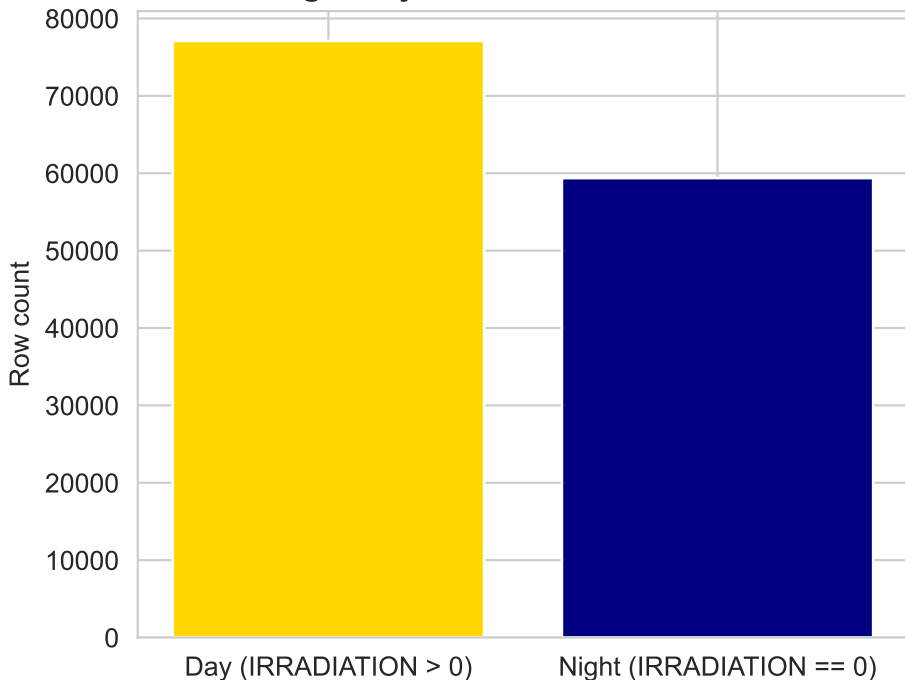
Mean AC_POWER per Inverter (SOURCE_KEY)



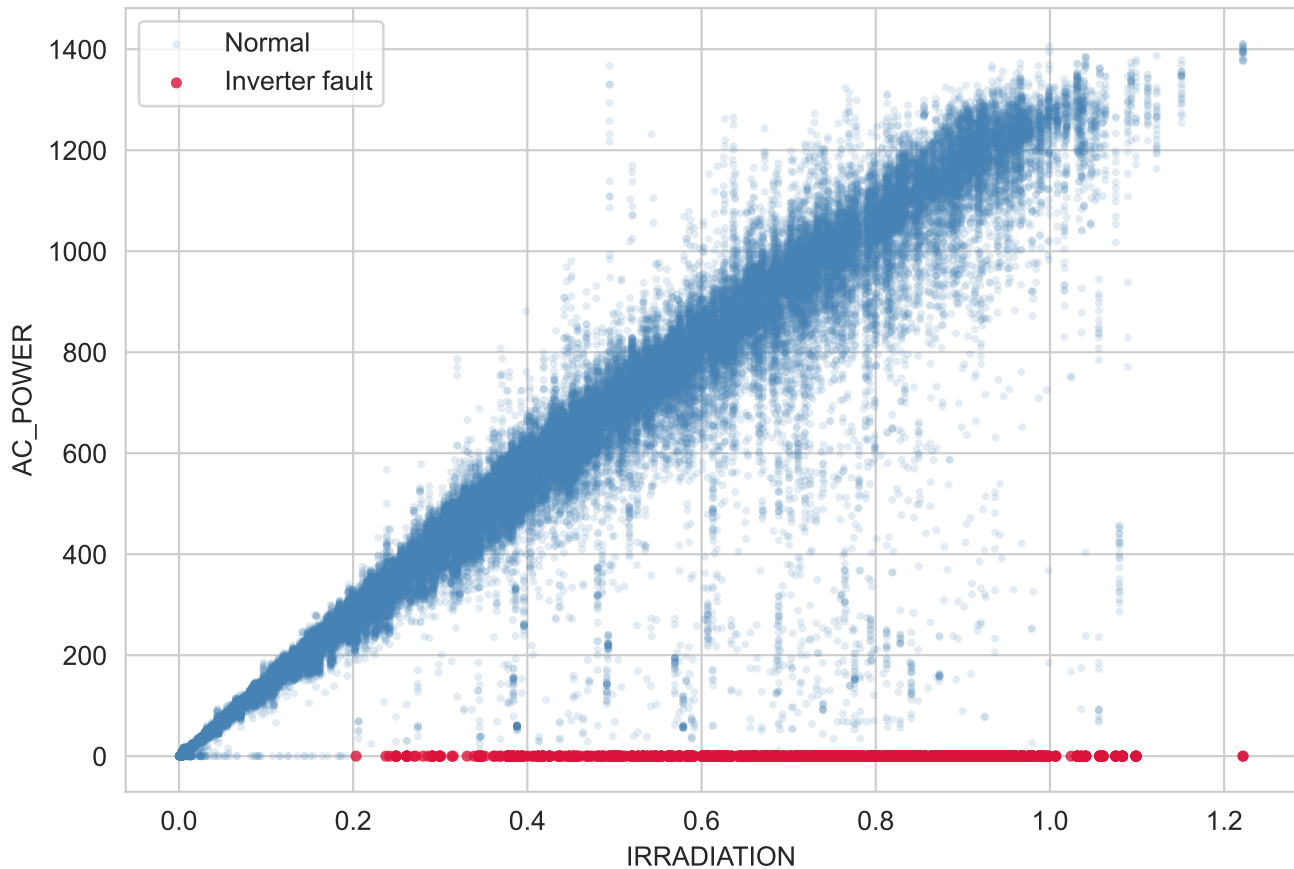
Autocorrelation of AC_POWER (inverter LIT2YUhhzqhg5Sw, up to 48h of 15-min lags)



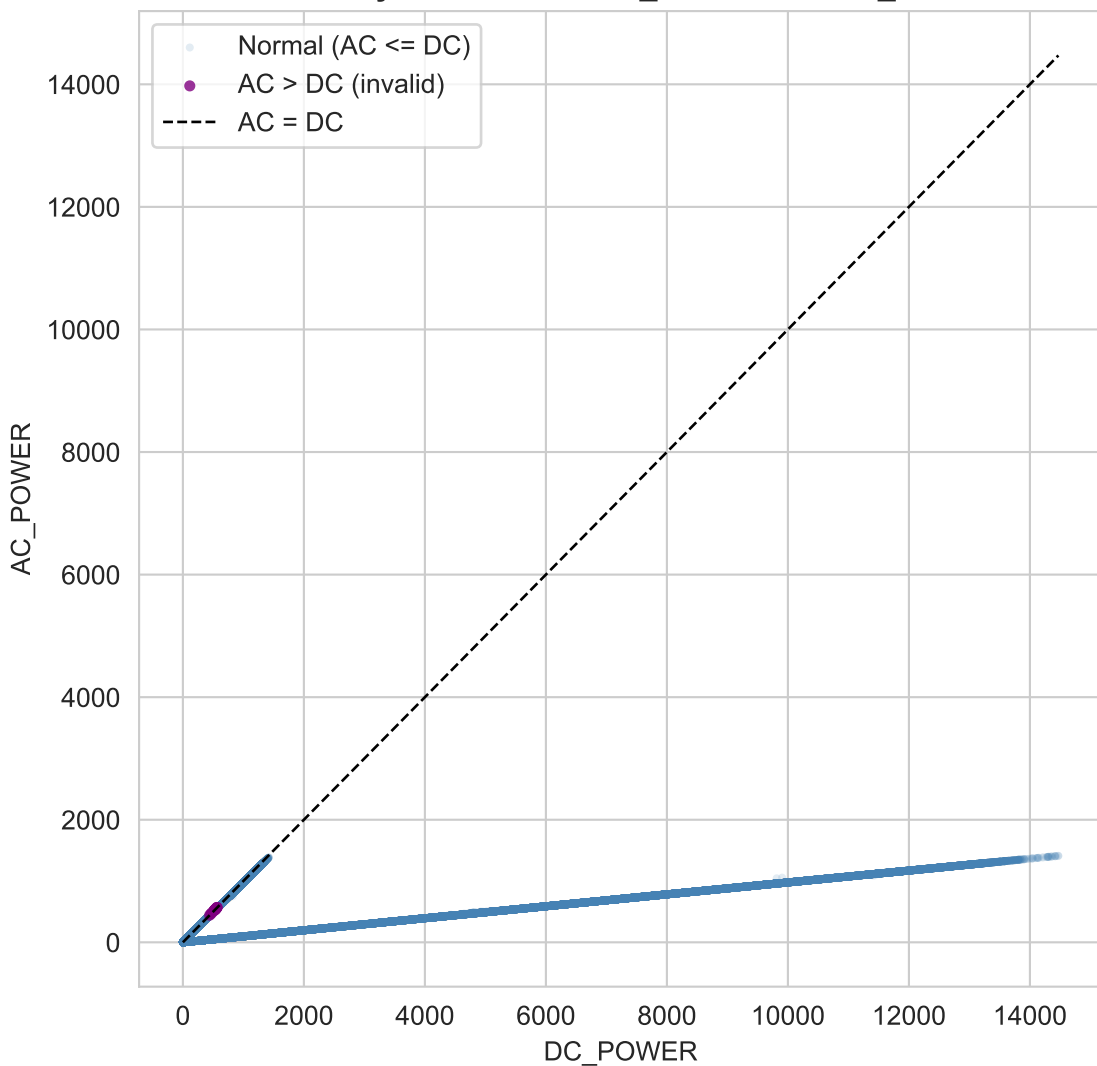
Night Adjustment: Rows Affected



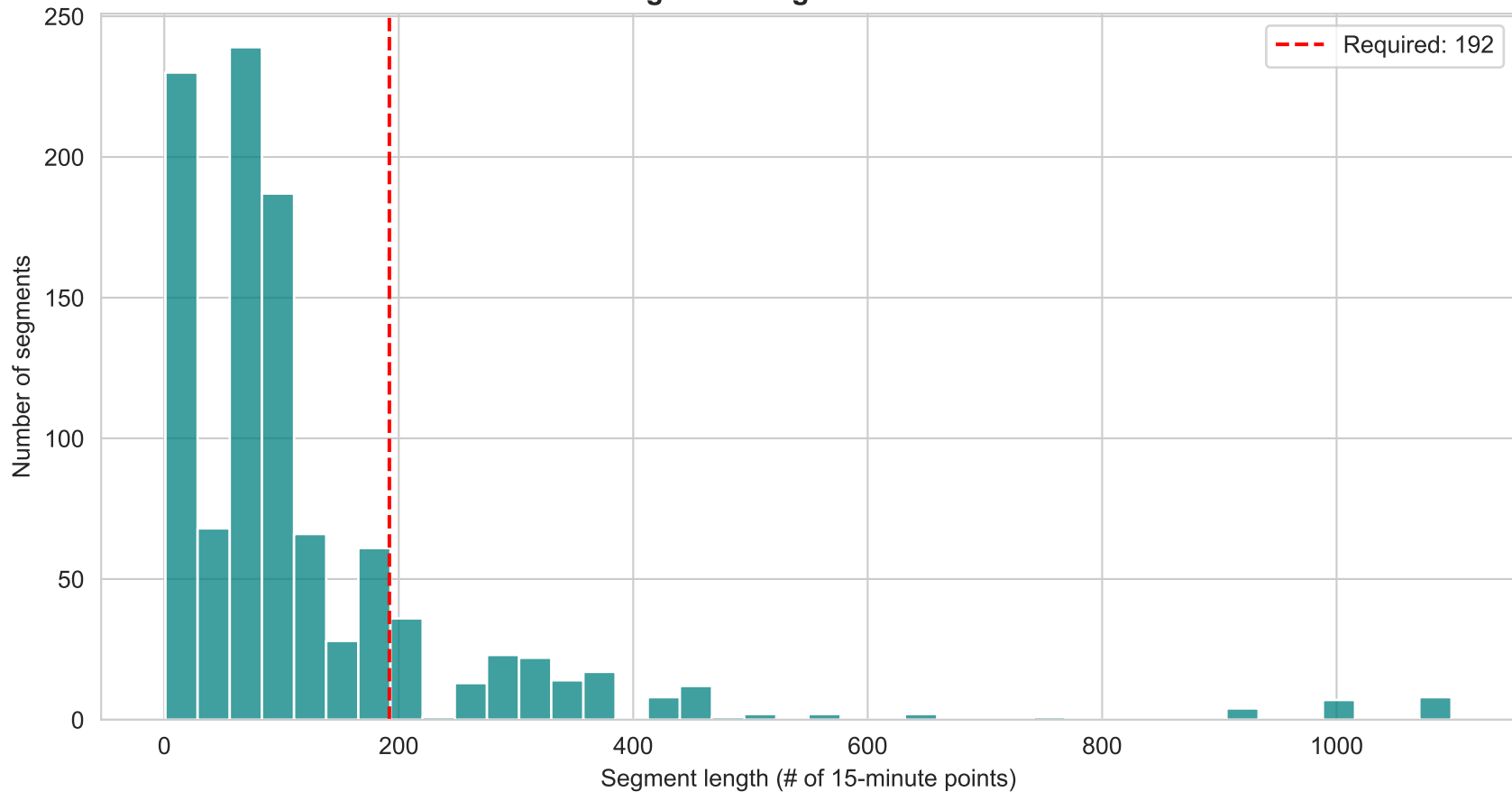
Fault Isolation: IRRADIATION vs AC_POWER



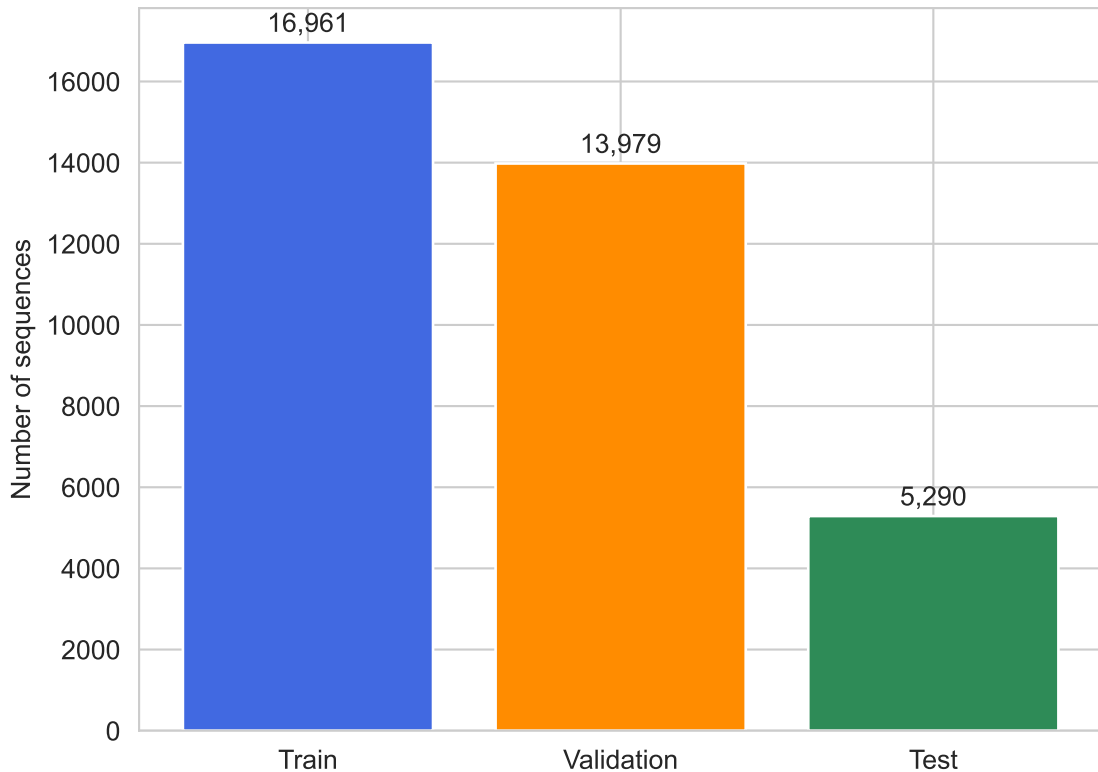
Invalid Physical Values: AC_POWER vs DC_POWER



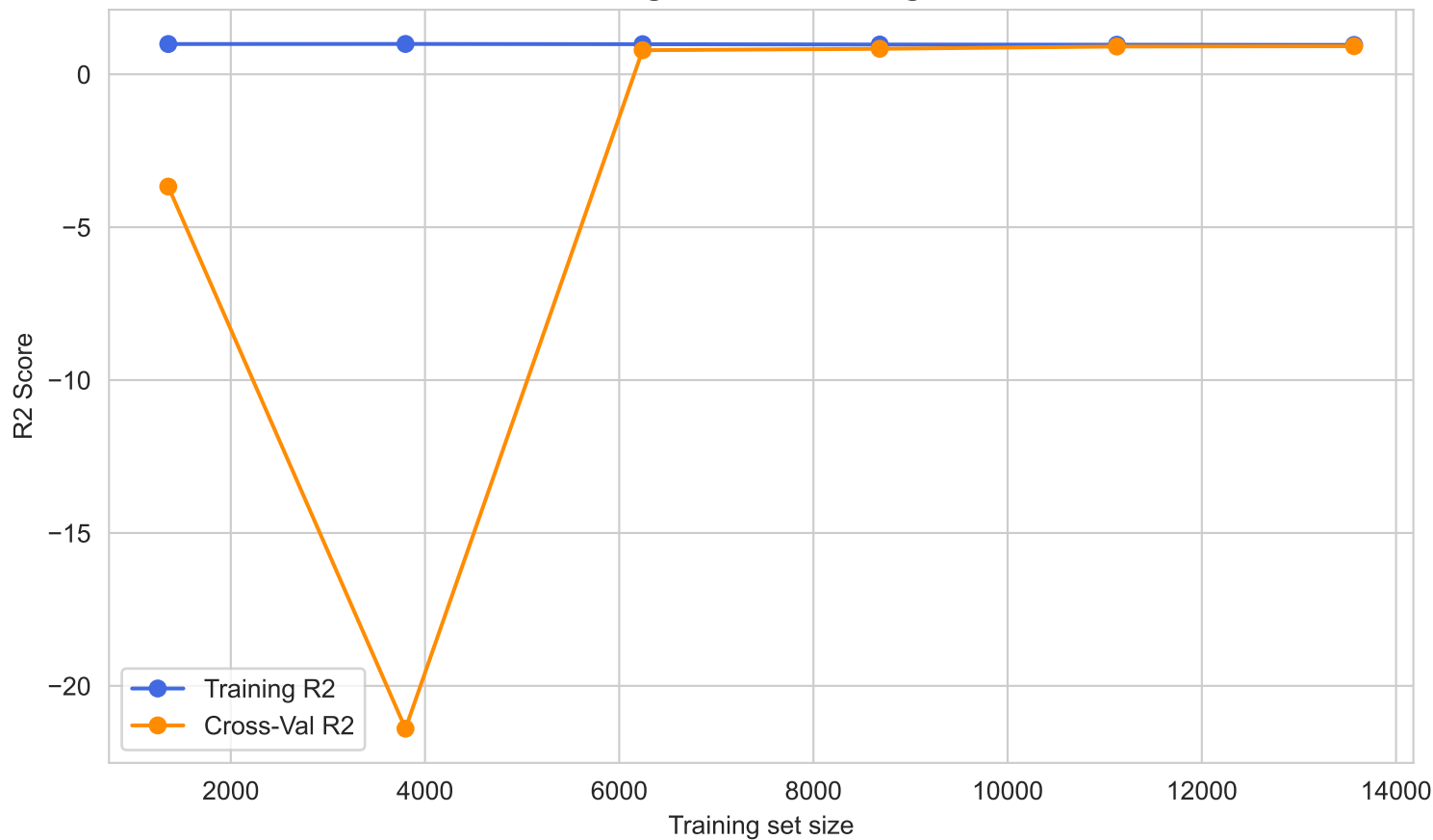
Segment Length Distribution



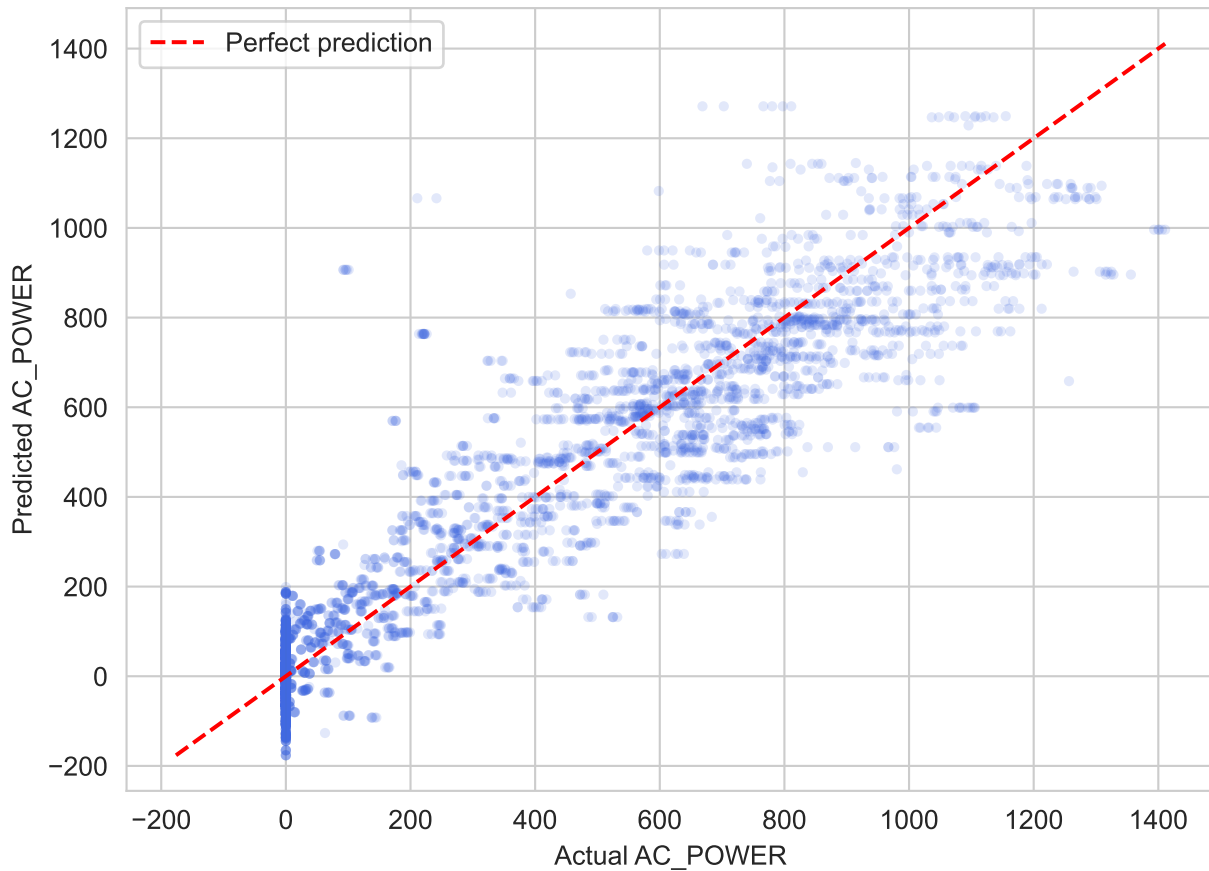
Sequence Count by Split (Chronologically-Ordered Eligible Segments)



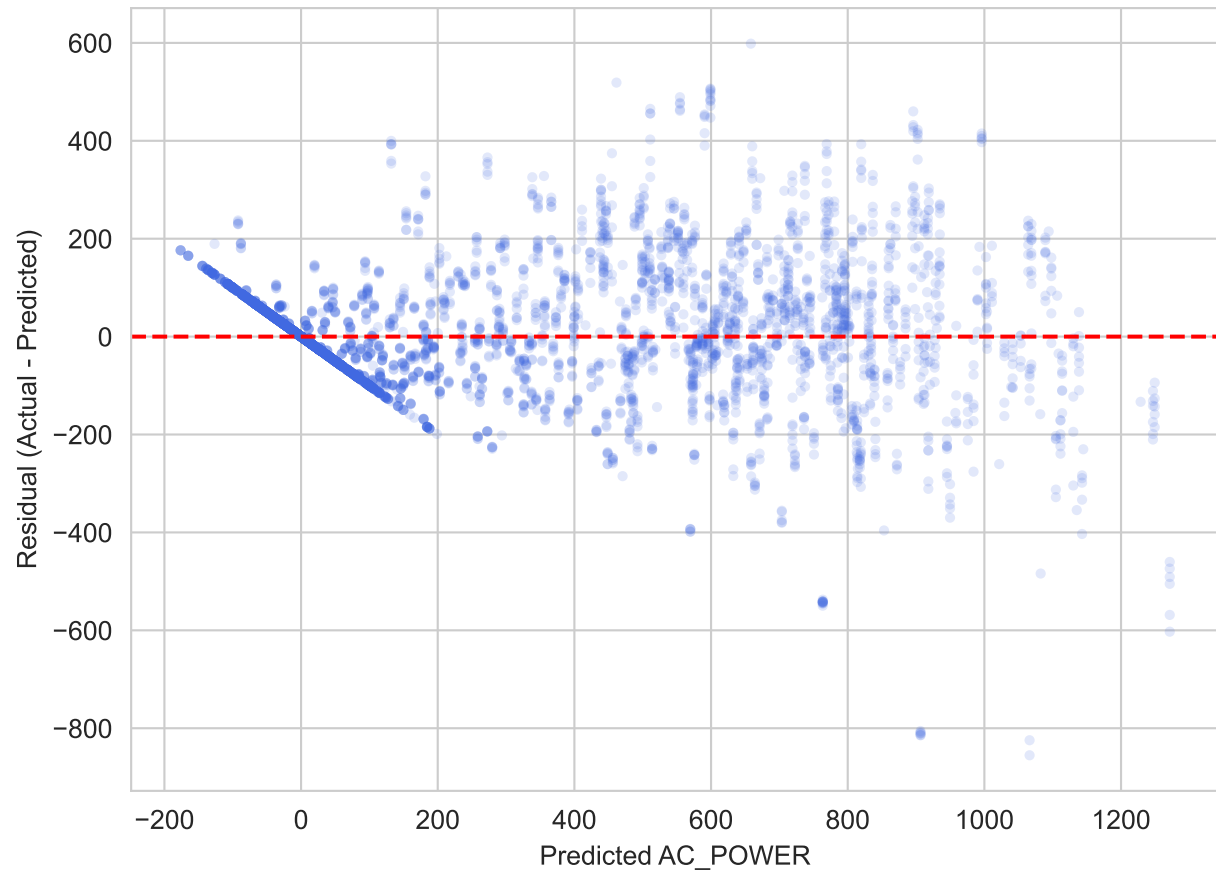
Linear Regression: Learning Curve



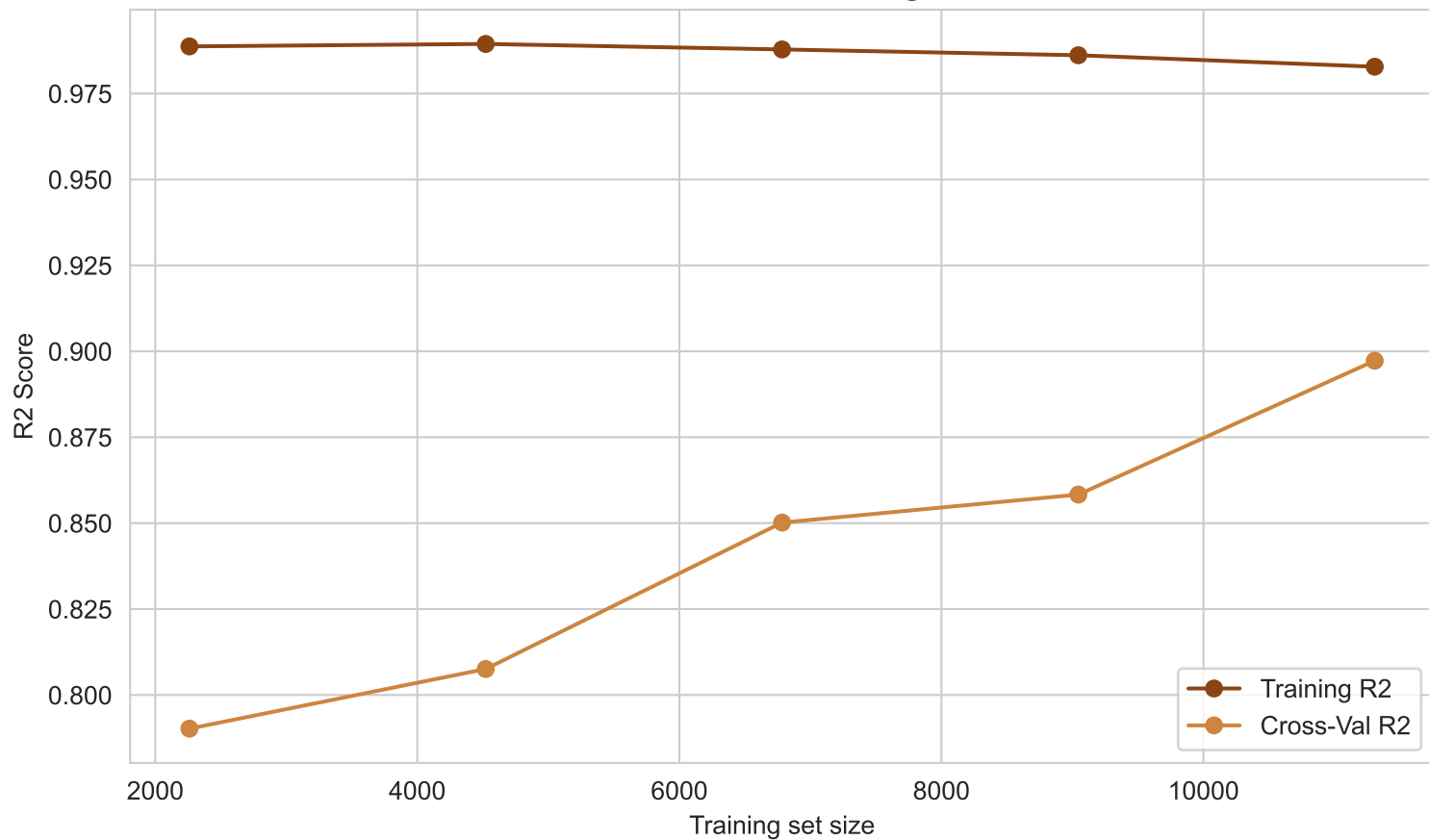
Linear Regression: Actual vs Predicted (Test Set)



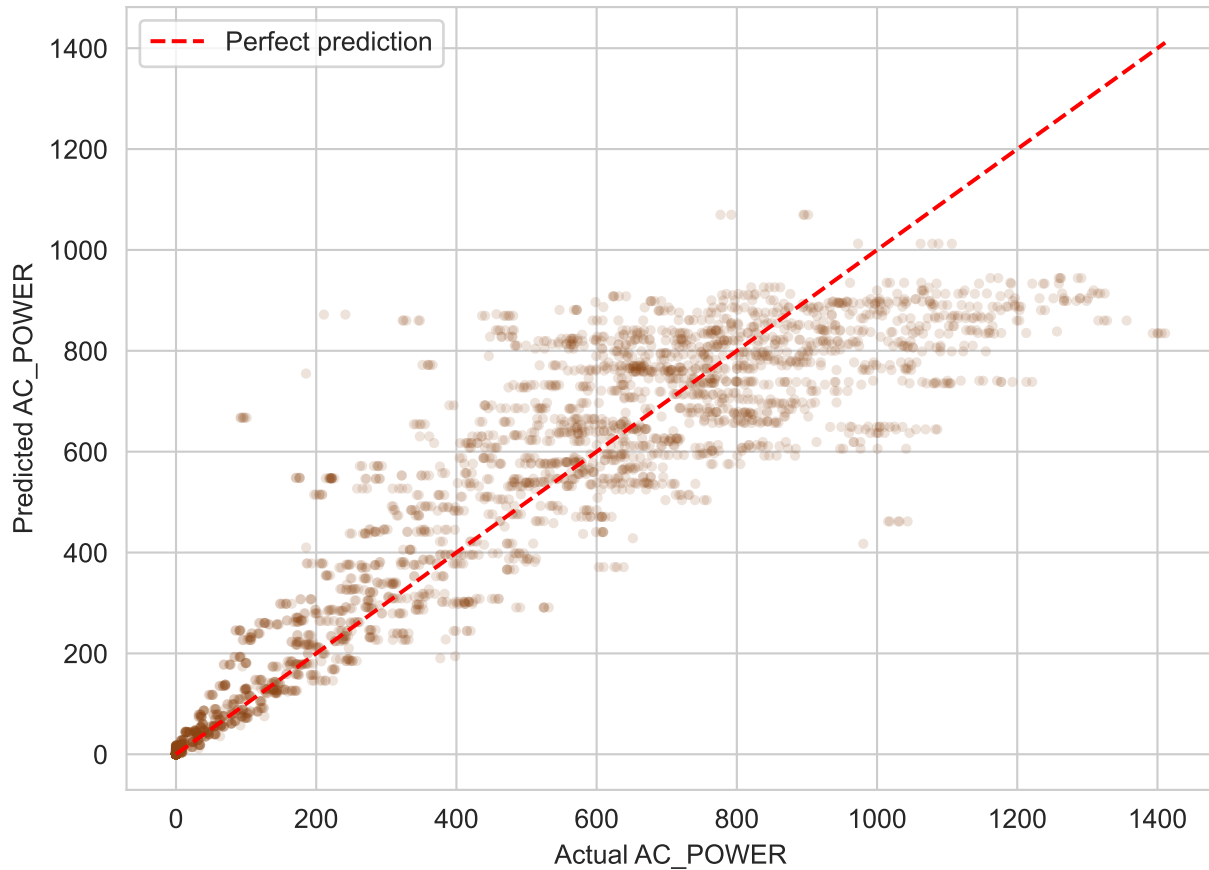
Linear Regression: Residual Plot (Test Set)



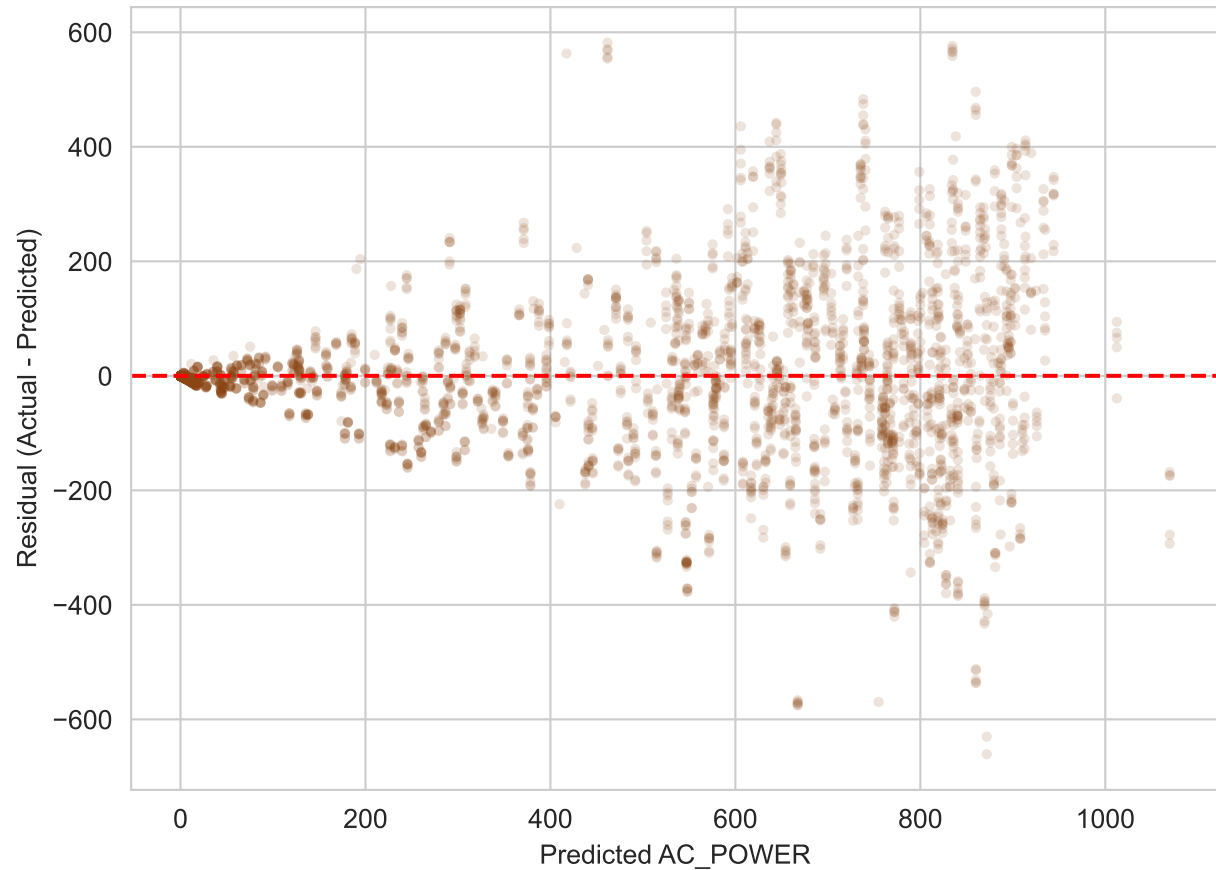
Random Forest: Learning Curve



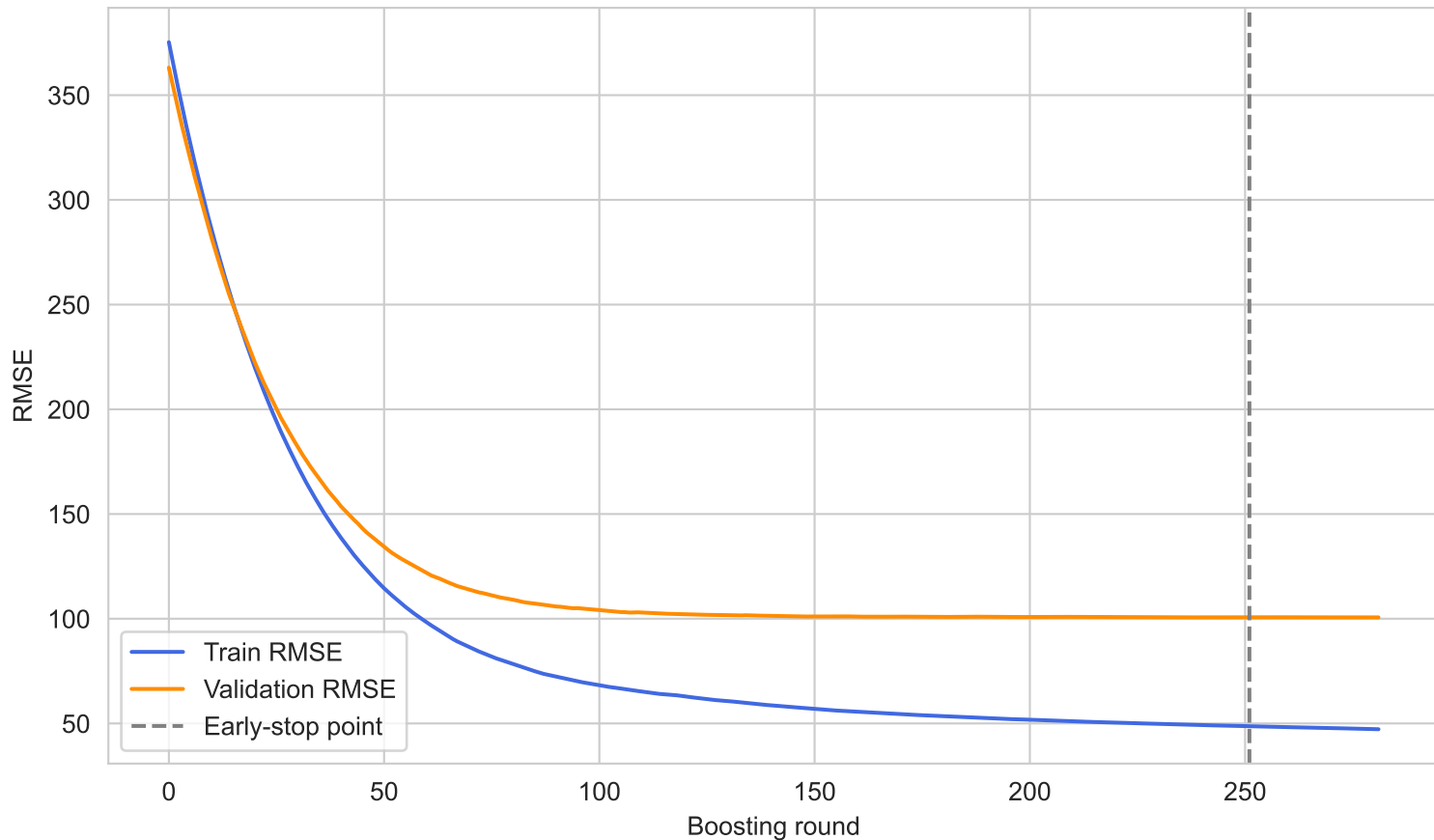
Random Forest: Actual vs Predicted (Test Set)



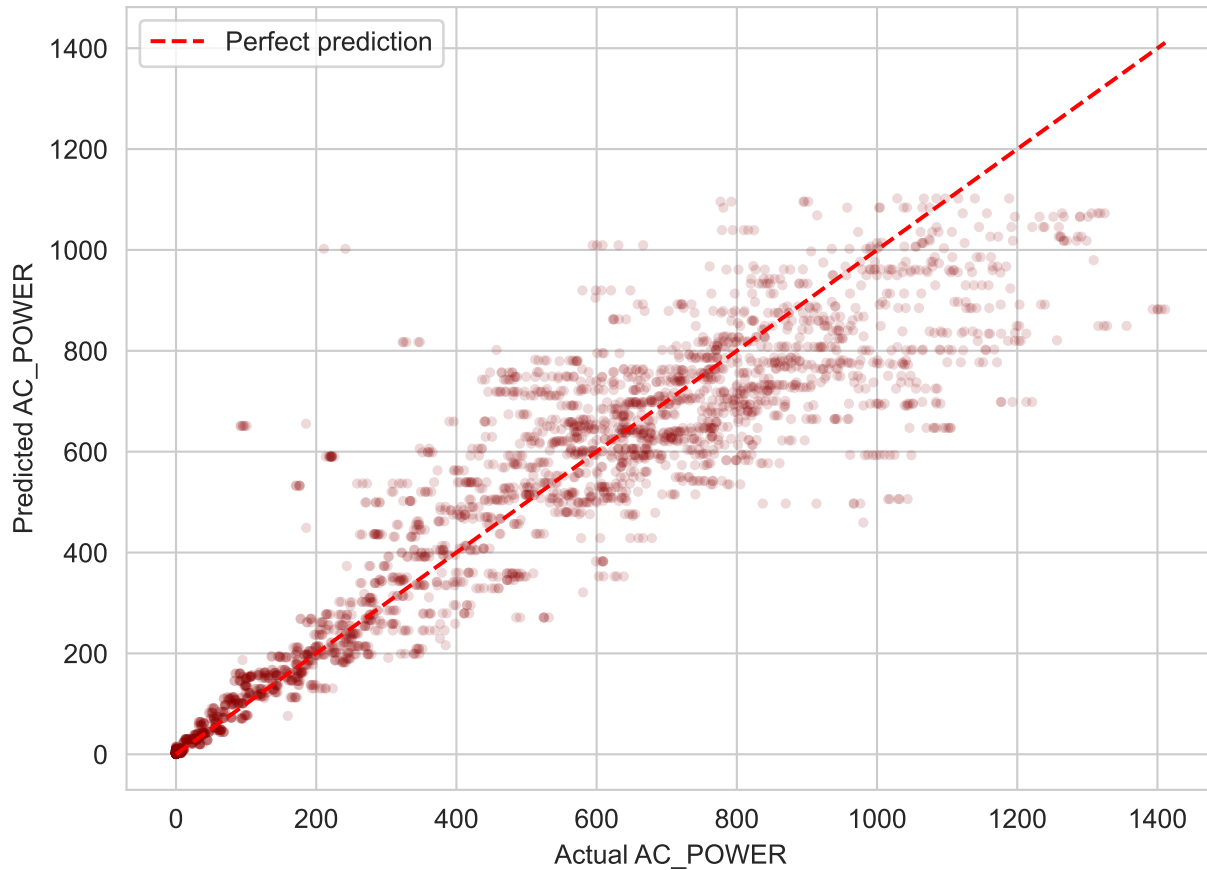
Random Forest: Residual Plot (Test Set)



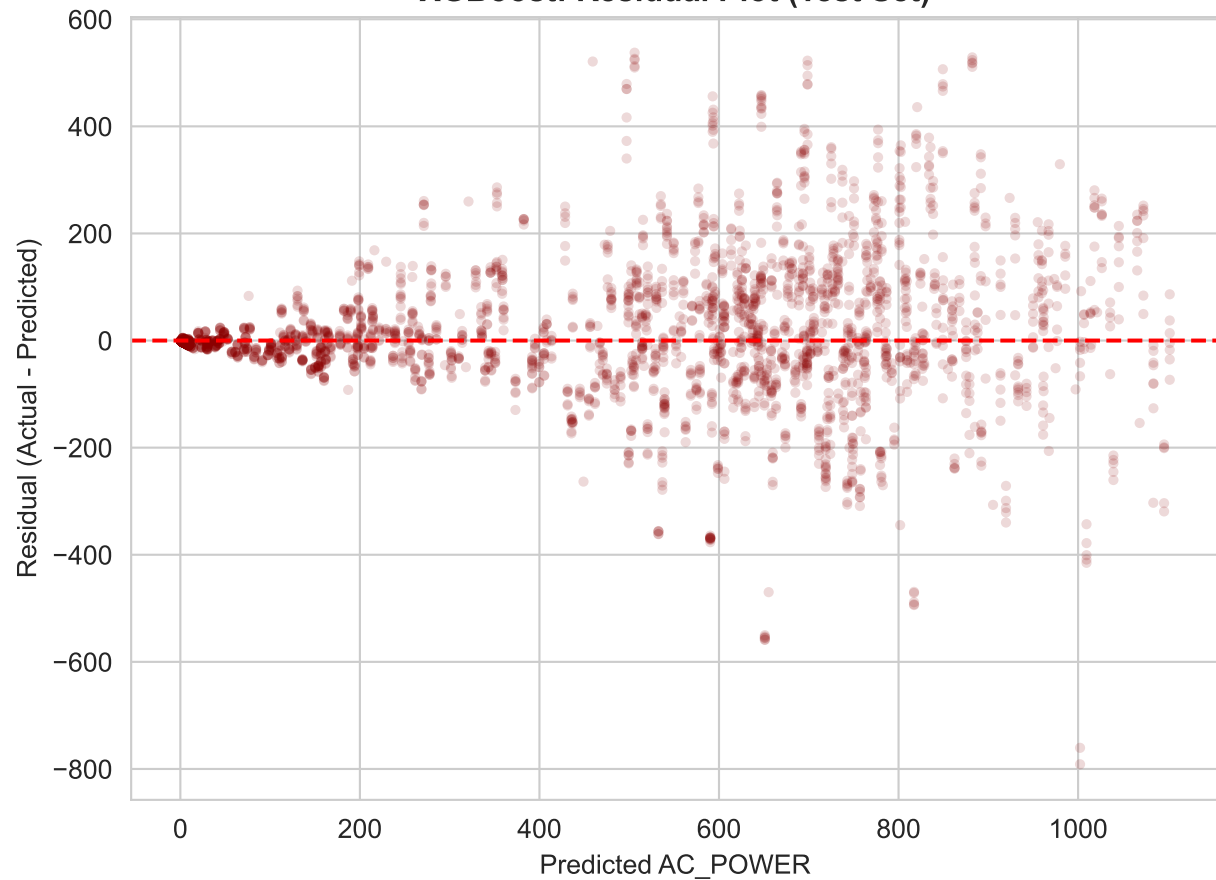
XGBoost: Learning Curve (Boosting Rounds)



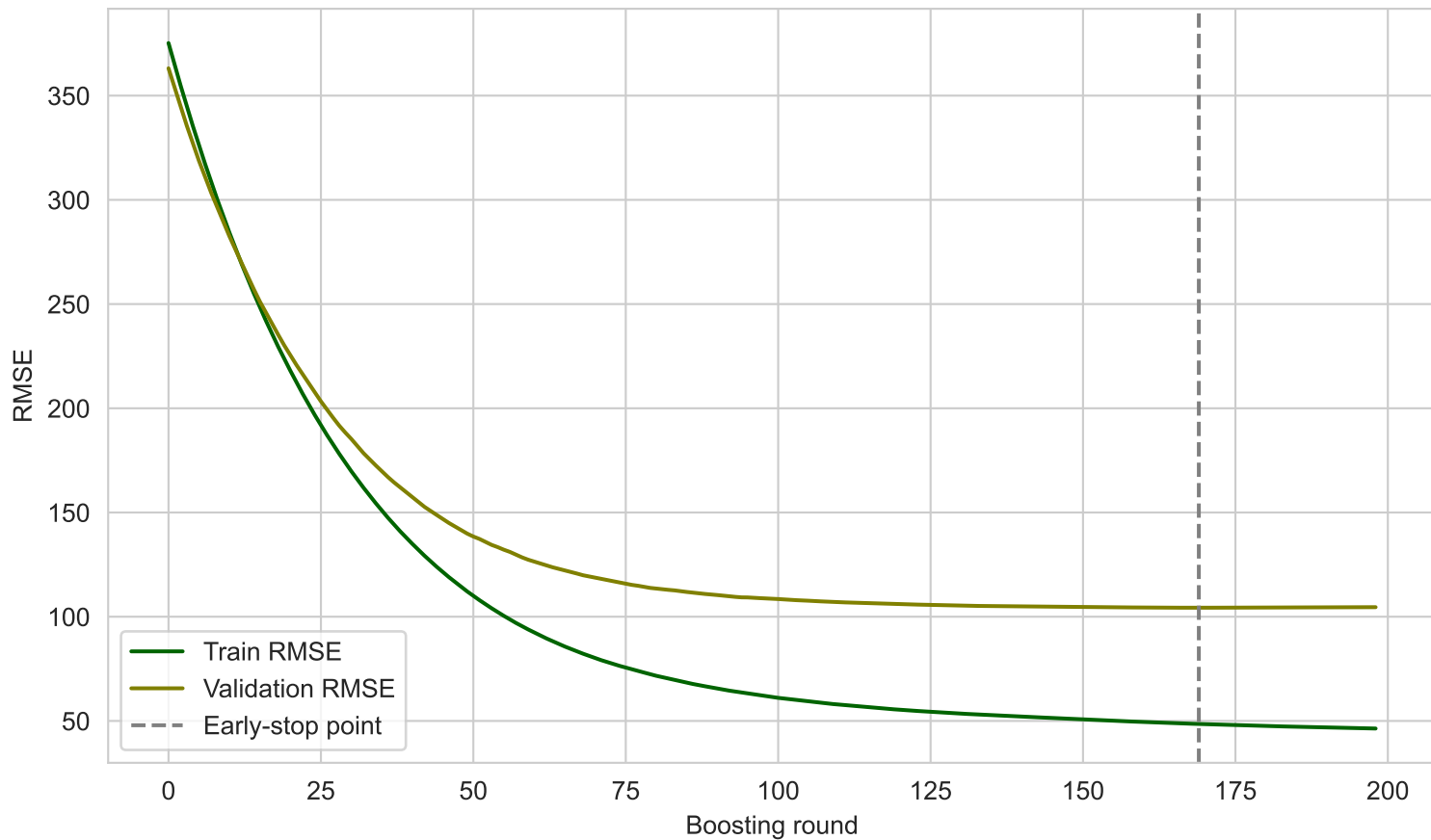
XGBoost: Actual vs Predicted (Test Set)



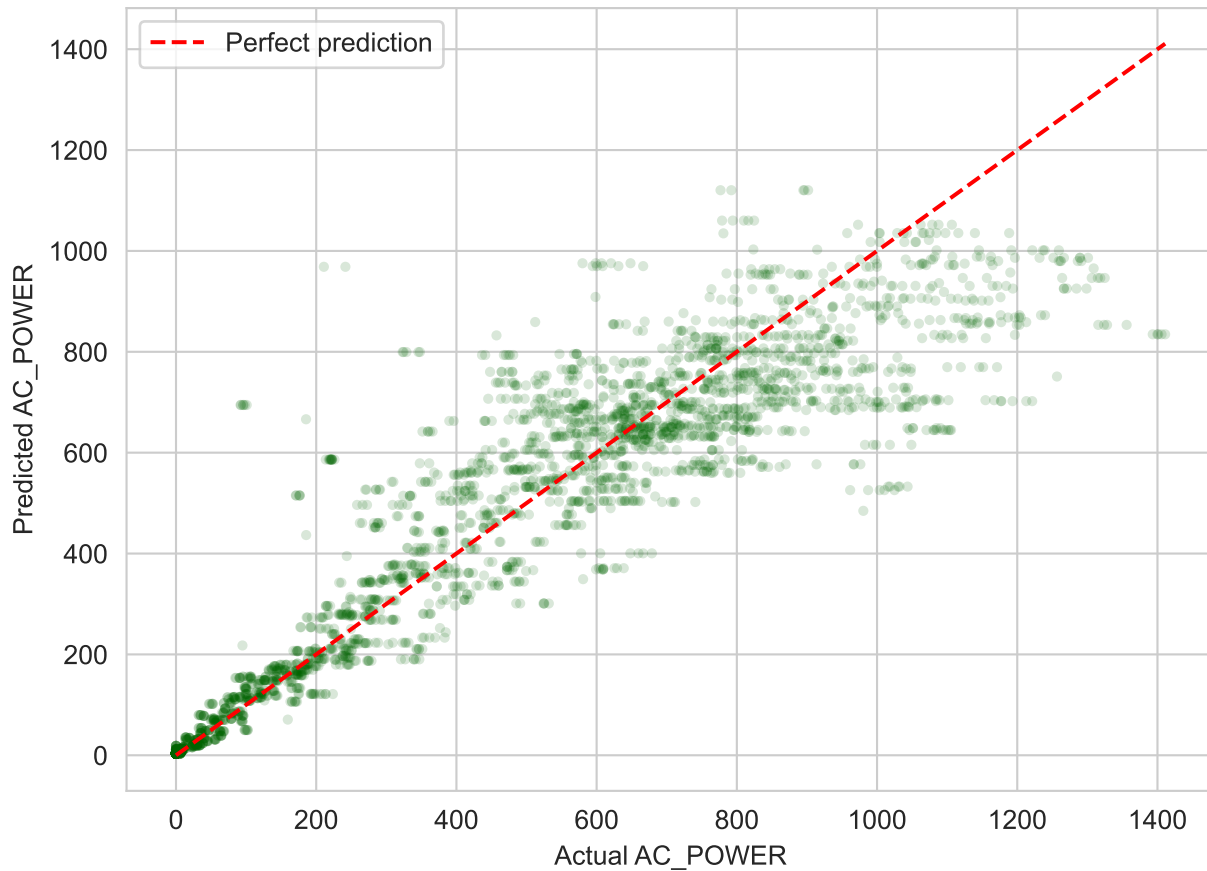
XGBoost: Residual Plot (Test Set)



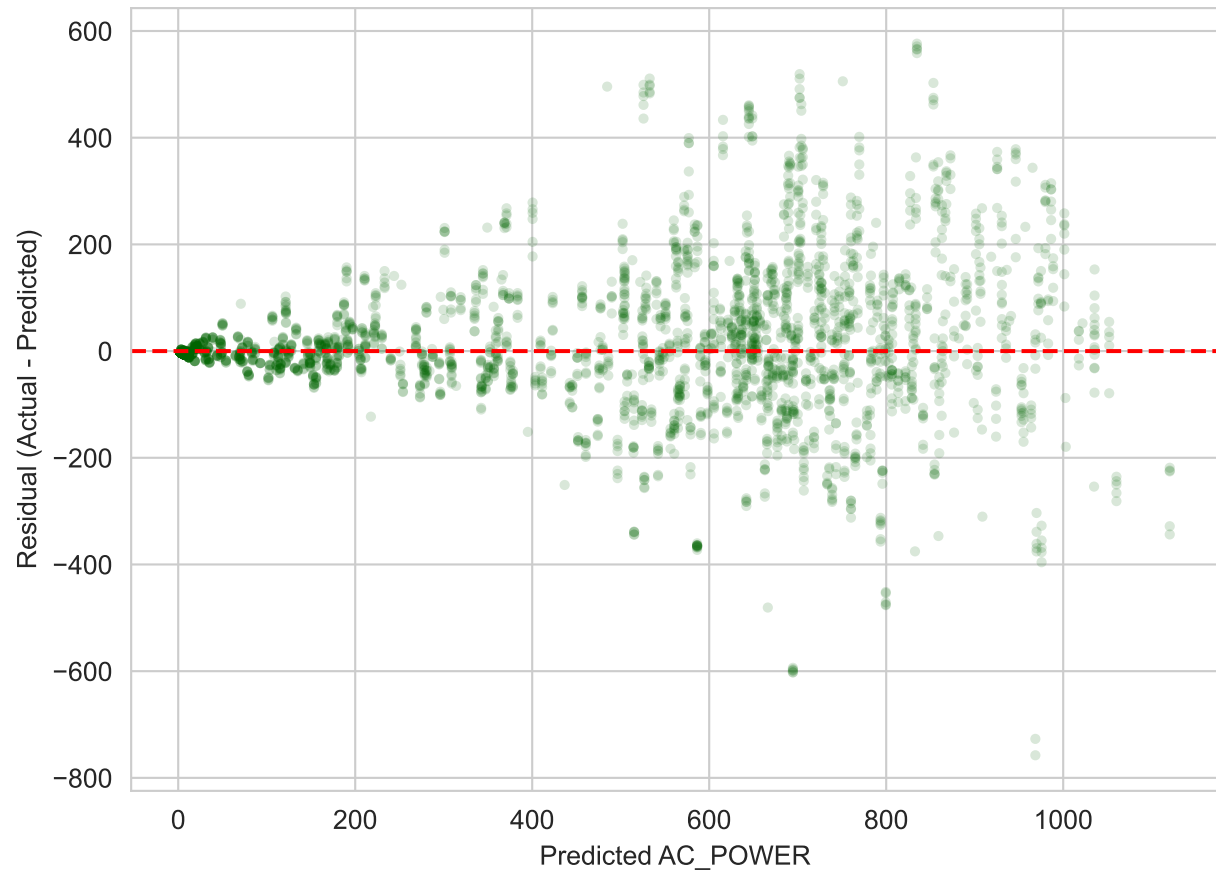
LightGBM: Learning Curve (Boosting Rounds)



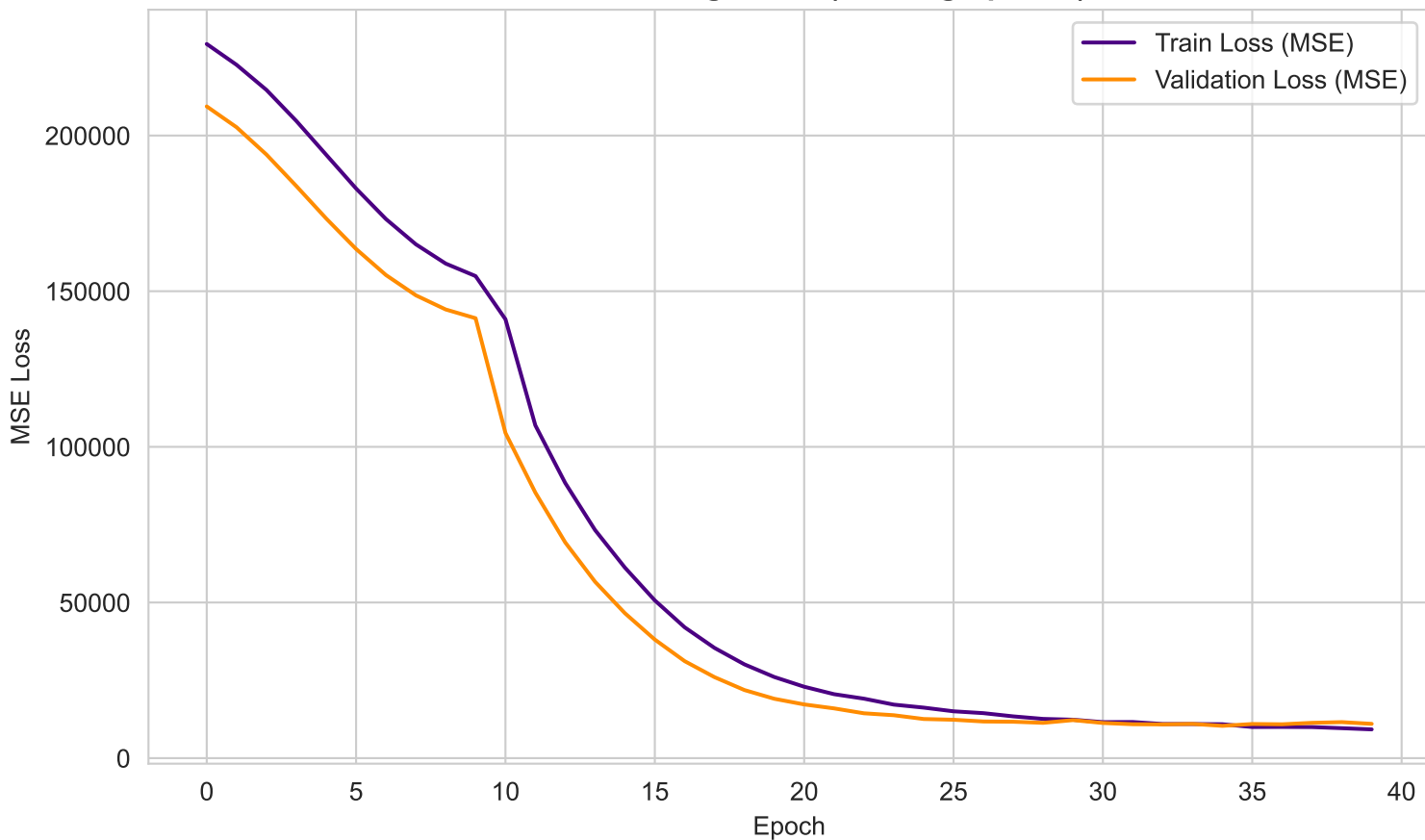
LightGBM: Actual vs Predicted (Test Set)



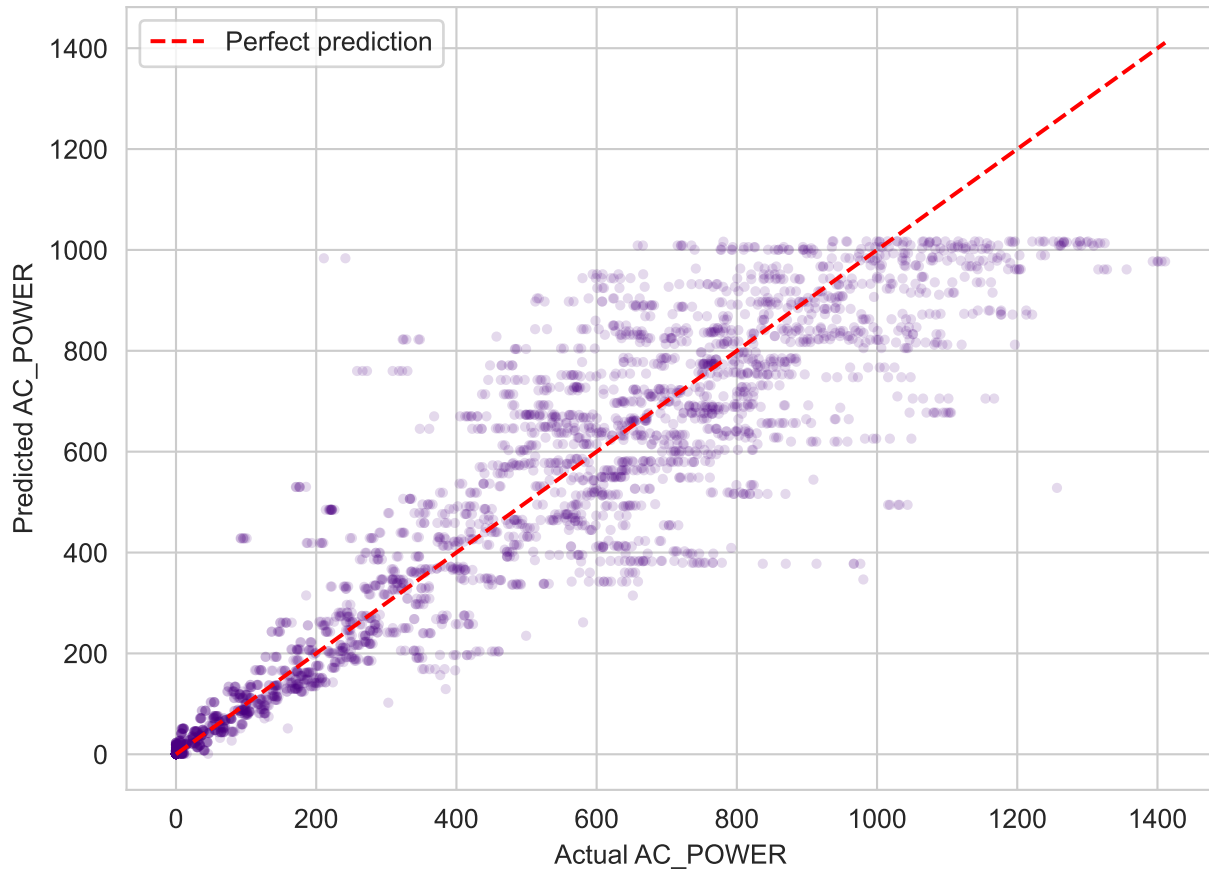
LightGBM: Residual Plot (Test Set)



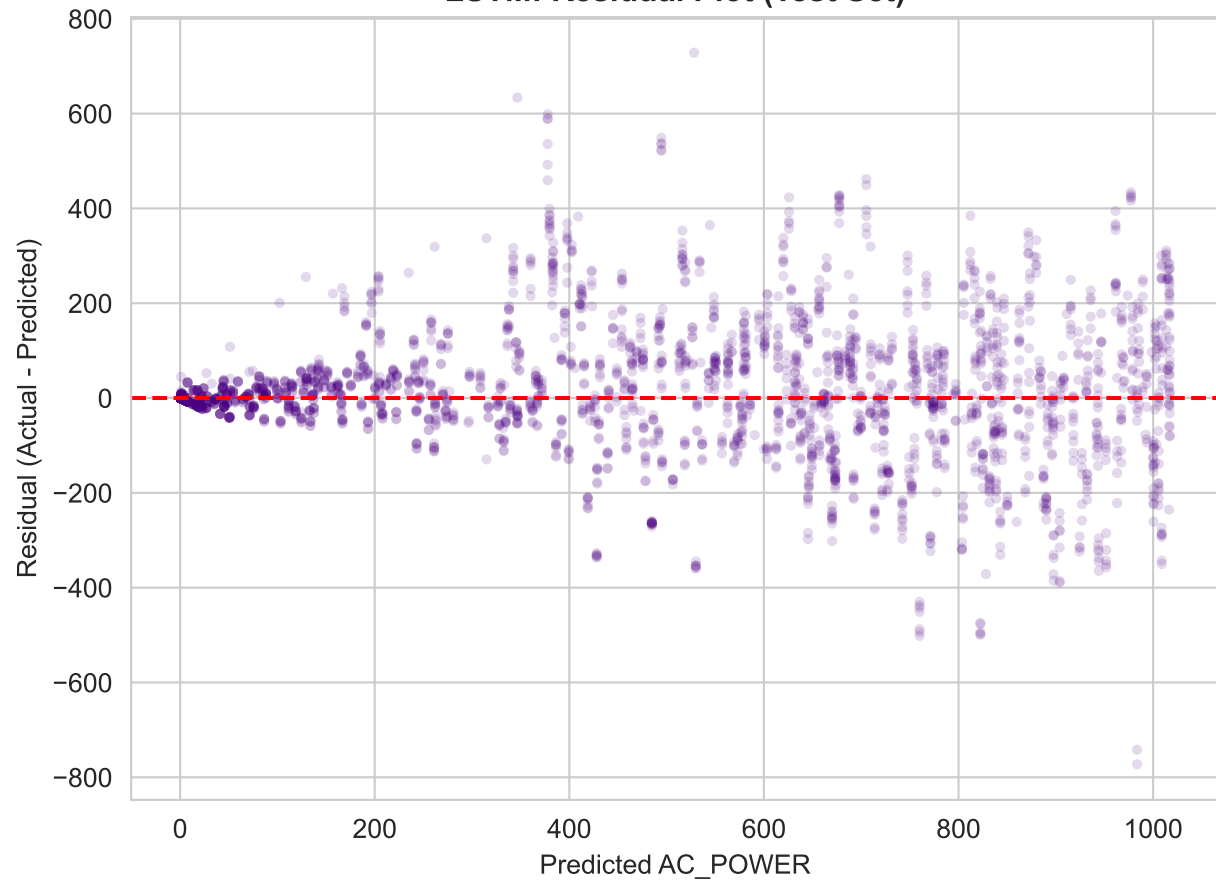
LSTM: Learning Curve (Training Epochs)



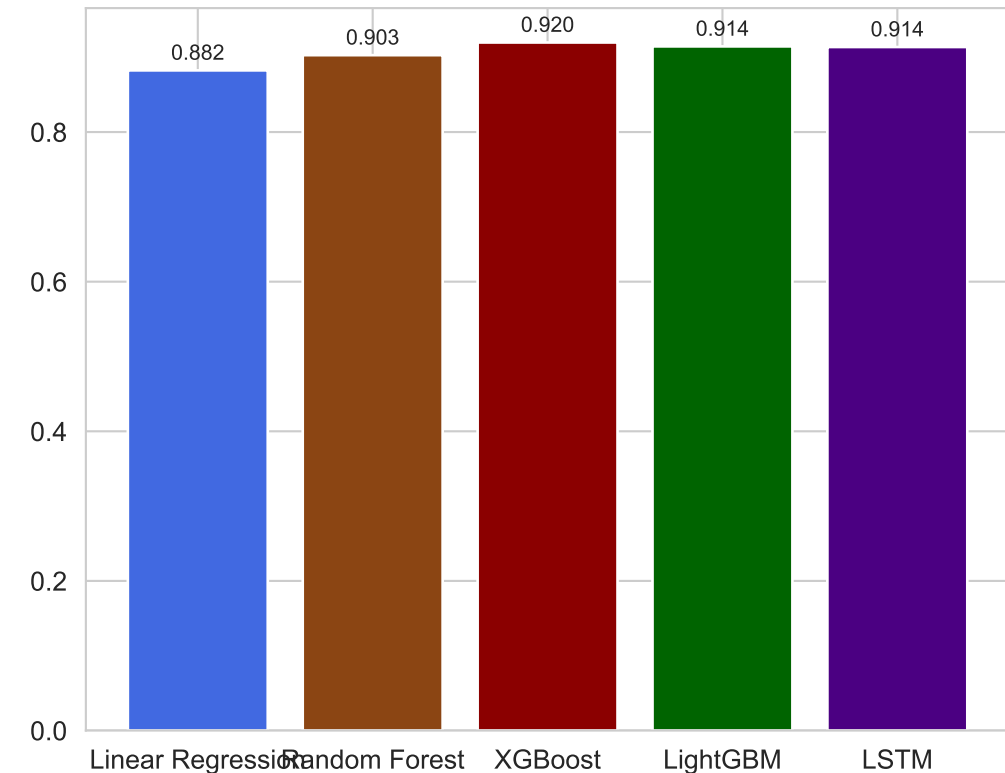
LSTM: Actual vs Predicted (Test Set)



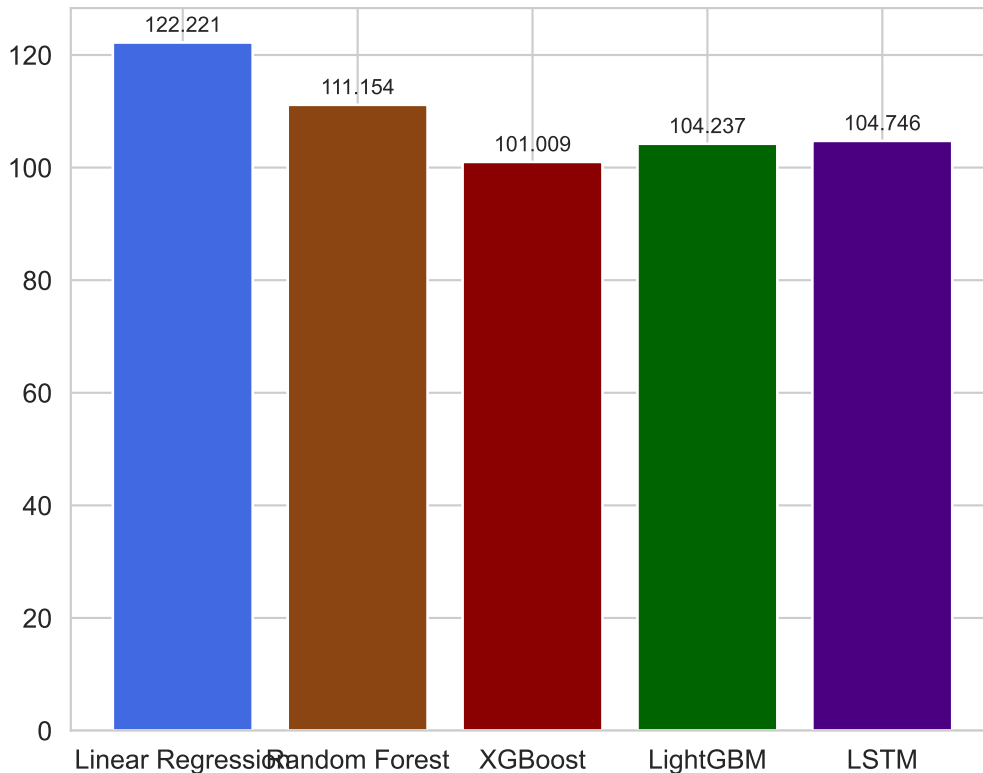
LSTM: Residual Plot (Test Set)



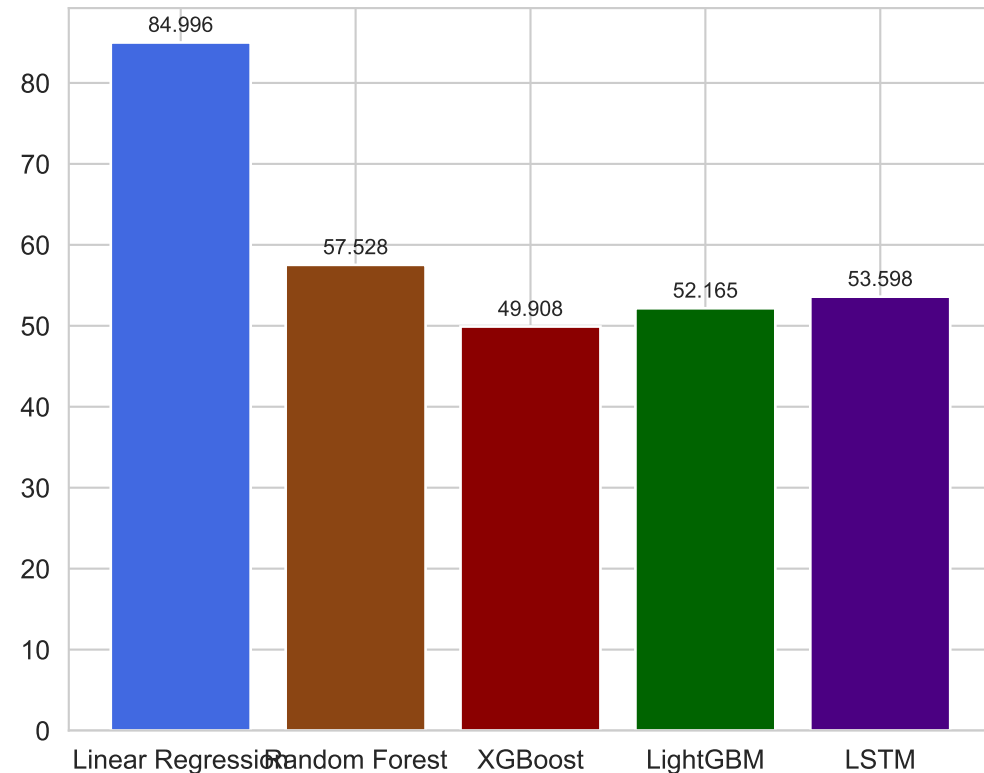
Test R2 Comparison



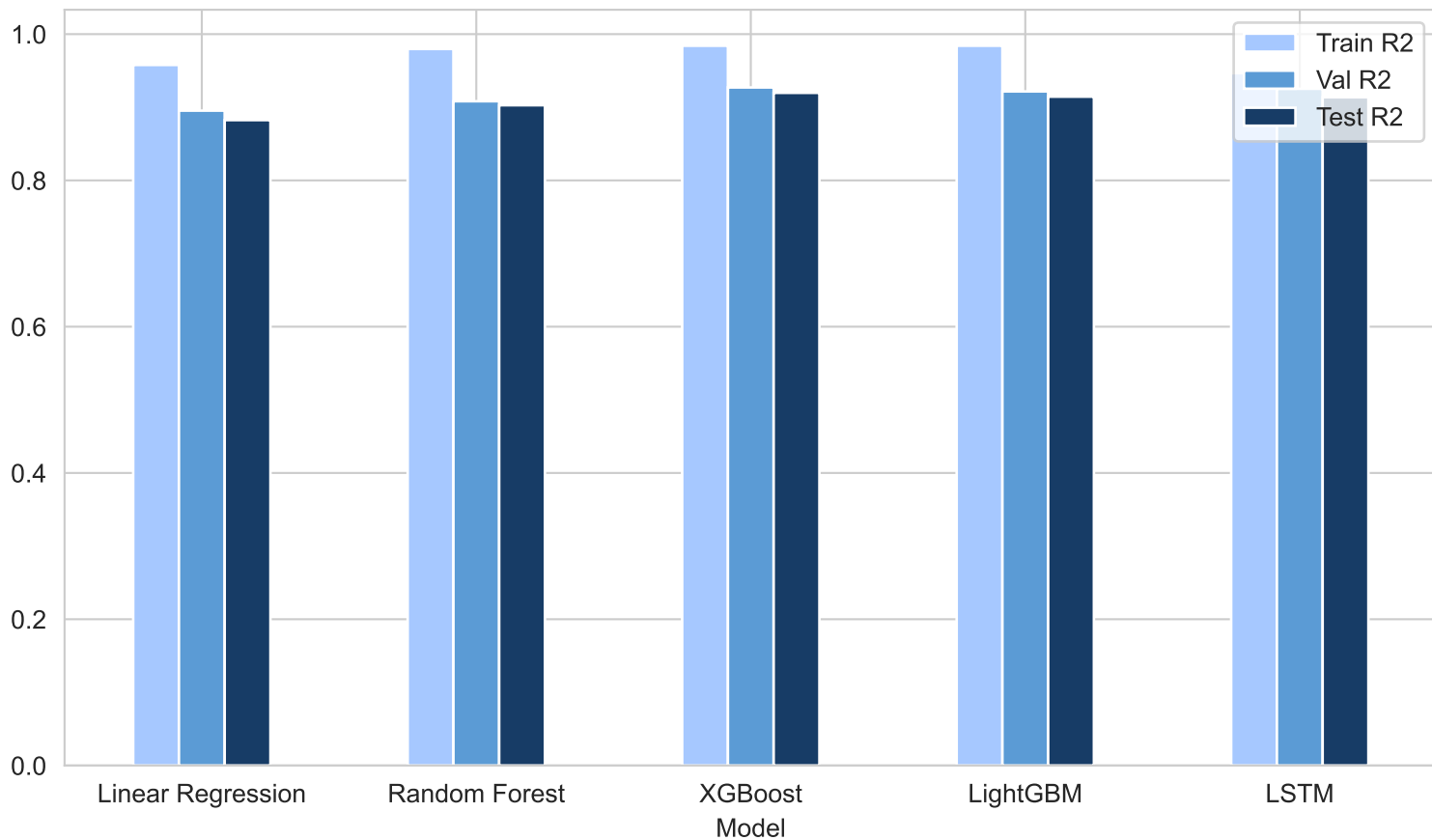
Test RMSE Comparison



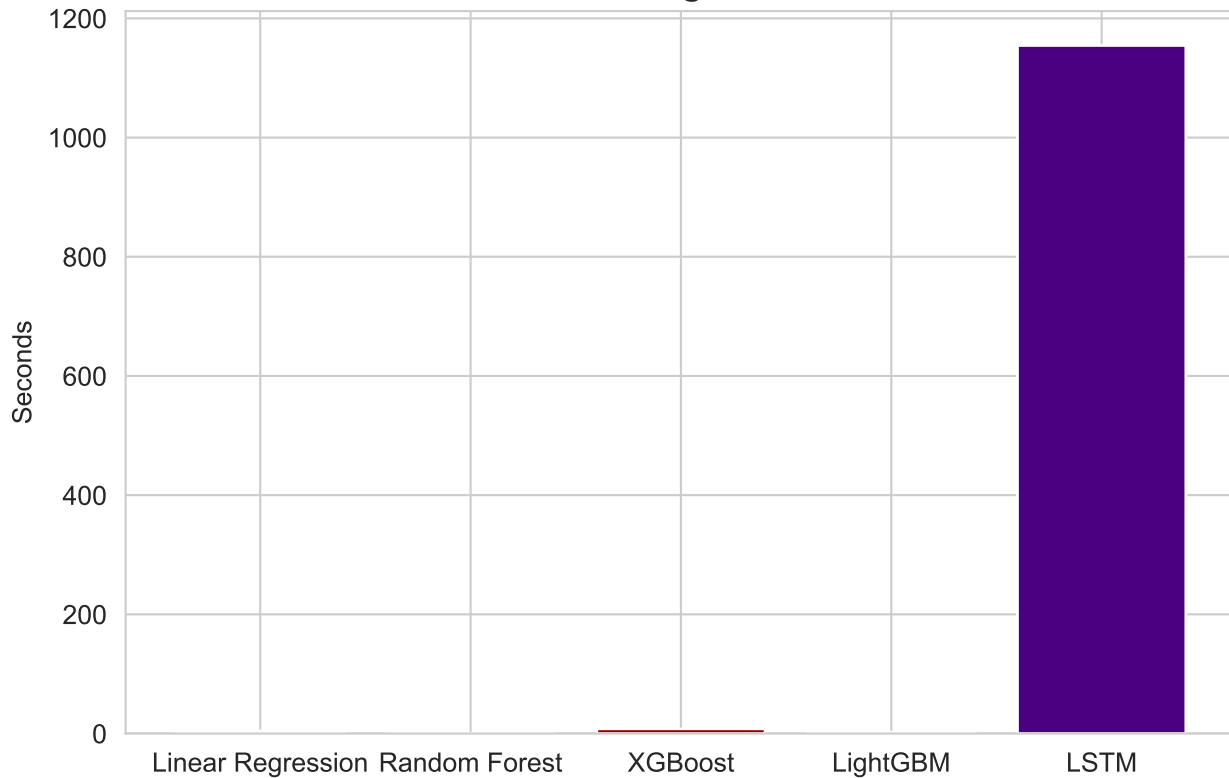
Test MAE Comparison



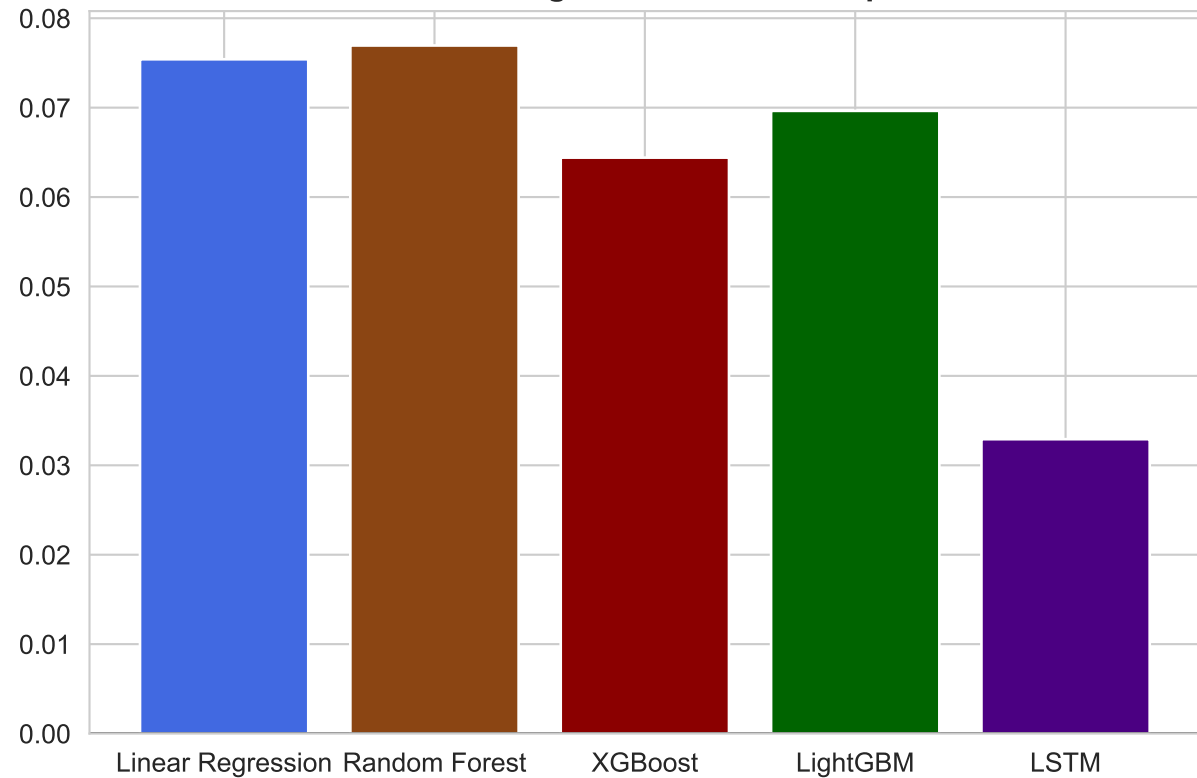
R2 Across Train / Validation / Test



Training Time



Overfitting / Generalization Gap



Model Comparison Table

Model	Train R2	Val R2	Test R2	Test MAE	Test MSE	Test RMSE	Training Time (s)	Prediction Time (s)	Overfitting Gap (Train R2 - Test R2)
Linear Regression	0.95785	0.89541	0.88243	84.99564	14937.8916	122.22067	4.39405	0.0	0.07541
Random Forest	0.97971	0.90824	0.90276	57.52807	12355.23984	111.15413	0.53975	0.01995	0.07695
XGBoost	0.98411	0.92707	0.9197	49.9076	10202.81152	101.00897	7.86154	0.00751	0.06441
LightGBM	0.98413	0.92165	0.91449	52.16549	10865.42264	104.23734	2.49713	0.00851	0.06964
LSTM	0.94657	0.92526	0.91365	53.59822	10971.64648	104.74563	1154.49775	2.36671	0.03292